

Interdisciplinary Studies in Human-Centred Design
Algonquin College

Connecting Flights: Understanding Passenger Rights and Flight Disruption Experiences

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DSN 8020 – DESIGN THEORY II

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The research question:.....	5
Section 1: Background Literature Review Research and Initial Insights.....	5
Background.....	5
Major Themes in This Research.....	6
Delays:.....	6
Missed Flights:.....	8
Policies.....	12
Section 2: Research Methods and Research Results.....	13
Part I: Research Question and Methods.....	13
The research question.....	13
Objectives (Sub-questions).....	13
Part II: Research Results and Recommendations.....	14
Questionnaire Findings:.....	14
General Findings.....	16
Analyzing Relationships:.....	18
Interview Findings.....	21
Theme 1: Unclear communication.....	21
Theme 2: Compensation issues.....	22
Theme 3: Knowledge of air passenger protection rights.....	22
Theme 4: Reasons for delays.....	23
Theme 5: Future travel expectations.....	24
What we heard: SME Interviews.....	24
Academic SMEs:.....	24
Industry SMEs:.....	25
Key Insights.....	25
Insight 1.....	25
Insight 2.....	26
Insight 3.....	26
Insight 4.....	26
Limitations of result.....	27
Experience Principles.....	28
Principle 1: I want transparency about flight disruptions.....	28
So what?.....	28
How might we as researchers:.....	28
Principle 2: I want to know my rights and where I can find them.....	28
So what?.....	29
How might we as researchers.....	29
Principle 3: I want it to be simple for me to understand.....	29
So what?.....	29
How might we as researchers.....	29

Principle 4: I want to know if I am eligible for compensation and how to receive it.....	29
So what?.....	30
How might we as researchers.....	30
Section 3: Solution Development, Testing, and Implementation.....	30
Personas.....	30
1. Priya.....	30
2. Jose.....	31
Journey Map.....	32
Mock-up (Storyboard).....	33
Validation.....	37
Testing Protocols.....	37
Feedback from Validation.....	37
Implemented Feedback.....	38
Final Prototype.....	38
Awareness:.....	38
Website:.....	39
Simplified APPR Guidelines.....	39
Compensation Eligibility Checker:.....	40
Reasons for Website Solution.....	42
Short-Term, Medium, and Long-Term Goals.....	42
What are the benefits for the client and the users?.....	43
Conclusion.....	43
Supporting Documents for Case Study Report.....	45
Appendix A: Full reference list for literature review.....	45
APPENDIX A1.....	60
Appendix B: Surveys.....	62
Appendix C – Subject Matter Expert (SME) interview materials.....	68
Transcription notes.....	68
Appendix D – Study participant interview materials.....	69
Participant recruitment protocol.....	69
Transcription notes:.....	69
Thematic analysis:.....	69
Appendix E: Research Design Artifacts.....	71
Personas.....	71
Journey Map.....	72
System Map.....	72
Storyboard.....	73
Initial Mock-up.....	74
Appendix F: Design Validation Materials.....	74
Validation Interviews and Protocol.....	74
Final Prototype.....	77

The research question:

What are the factors contributing to the communication gap between the services provided to passengers and their expectations during disrupted connecting flights, particularly concerning passenger rights and compensation?

Section 1: Background Literature Review Research and Initial Insights

Background

The number of elements that cause air travel cancellations and delays is such a huge challenge for the aviation industry. In the context of a rapidly changing travel industry, including more travellers and changes in their travel behaviour, getting to the bottom of the multifaceted nature of these risks is a top priority.

According to Cattaneo et al. (2019), approximately 40.3% of connections in Europe resulted in passengers missing their outgoing flights to their final destinations. This occurred when the incoming flight from the origin airport was delayed by 30 minutes at the hub. Such disruptions create financial burdens such as rescheduling fees or additional accommodation costs (depending on the airline's policies) and evoke negative emotional responses from passengers, often resulting in adverse word-of-mouth sentiments (Lawson, 2023; Jiang et al., 2019). Despite these challenges, passengers are often unaware of their rights when it comes to compensation options for flight delays and cancellations (Geng, 2023). Airlines offer compensation for flight disruptions and training staff for quality services, yet a significant gap exists in passengers' awareness of their right to compensation. This suggests airlines might not be effectively communicating these rights to passengers, leaving them uninformed until they experience any flight disruptions (Toolkit, 2023; Legislative Services Branch, 2023). It is important to address the communication gap between airlines and passengers because a lot of travellers don't know their rights, which might lead to missing out on getting compensation for the flight disruption they faced (The Canadian Press, 2019).

This research focuses on several key objectives. It aims to identify the main reasons for the communication gap between the services provided to passengers and their expectations during missed connecting flights. Additionally, it evaluates whether airlines make it clear and easy for passengers to understand their entitlements when disruptions occur. The research examines how proactive communication and customer engagement techniques can build trust and loyalty among passengers. It also analyzes how passengers' experiences differ from their expectations during missed connections.

Major Themes in This Research

Delays:

Flight delays can have a profound impact on airline operations as well as passengers. An understanding of the complexity of these delays is necessary for assessing their wider consequences for the travel industry. It is one of the major areas of concern for airline operators, the airport, and the passengers. Understanding how a delay happens and its impact is critical to examining the effects on the whole travel industry. Below are the sources and causes of delays that will be looked at in this study.

Sources and causes of delays:

Understanding the elements that contribute to delays is essential for the airline business. Many delays are short, but they can lead to longer delays, especially if many variables are involved (Henriques & Feiteira, 2018). As air traffic grows, the percentage of flights arriving on time decreases, highlighting the need to address this issue (Mitsokapas et al., 2021). Understanding the subtle factors behind flight delays is crucial for enhancing operational efficiency and increasing passenger satisfaction in the aviation sector.

A flight that is delayed by 15 minutes or more from its planned time is considered a delayed flight (Henriques & Feiteira, 2018). 70% of the delays are observed to be less than an hour (Mitsokapas et al., 2021). If a flight is delayed for two or more reasons, there is a greater than 30% probability that it will be delayed again on the same trip (Henriques & Feiteira, 2018). As the number of flights consistently increases annually, the rate of flights arriving on time decreases from 83.19% in 2007 to 71.67% in 2017. (Mitsokapas et al., 2021). Gate assignments, airport infrastructure limits, and scheduling problems have all been highlighted as important causes of delays, particularly at low-cost airports (Ali et al., 2019).

According to the US Bureau of Transportation Statistics, there are five categories of the causes of passenger flight delays. They have divided these categories into propagated delays and non-propagated delays. The non-propagation factors include the carrier, extreme weather, the national aviation system, and security. The factor of late-arriving aircraft is directly causing Propagated Delay (PD) (Bureau of Transportation Statistics, 2024). The departure delay of the current flight is inevitably caused by the late arrival of the immediately preceding flight with the same aircraft (Wang et al., 2020). The unpredictable weather conditions, wind, and military exercises have a significant impact on delays. Airline policies, flight routes, technical defects, or documentation issues contribute to 27% of all delays. (Mitsokapas et al., 2021). Flow control, the number of scheduled departures, technical failures, and route restrictions also have a crucial role in causing delays (Cheng et al., 2019). Delayed entries and technological malfunction are two of the most important issues (Cheng et al., 2019). In line with the propagating effect of flight

delays, the variables linked to prior flight delay information were found to have the greatest influence on whether the current flight is delayed or not (Fernandes et al., 2019).

Ground operations have a significant impact on how long layovers last and how well flight operations run. According to Kešel'ová and Hanák (2019), ground operations include maintenance, refuelling, and luggage handling. However, if faults or inefficiencies arise during these activities, they could lead to delays. As a result, such delays can lead the aircraft to spend excess time idle during layovers. Kešel'ová and Hanák (2019) underline the importance of improving communication and coordination between airlines and airport authorities to address these difficulties. Improving teamwork and communication channels can help to reduce airport delays. Furthermore, managing ground activities is essential for decreasing layover duration and increasing customer satisfaction (Kešel'ová and Hanák, 2019). By improving ground operations coordination and efficiency, airlines and airport authorities may efficiently streamline layover processes and improve overall flight operations (Kešel'ová and Hanák, 2019).

Problems with boarding can cause delays and make layovers more challenging. Van Landeghem and Beuselinck (2002) investigate how passenger boarding times influence flight punctuality and cause delays. Inefficient boarding processes can cause gate congestion and disorganization, resulting in increased flight delays and reduced passenger satisfaction (Van Landeghem and Beuselinck, 2002). Passengers may get confused and anxious when boarding areas are packed and the process is disorganized. Using more structured boarding procedures, such as zone-based boarding or new technology, can speed up the process and minimize wait times (Van Landeghem and Beuselinck, 2002).

Emotional influence of flight delays on passengers:

There is a complex relationship between airline service delays and passengers' emotional reactions, which reveals significant insights into the behaviors of affected travelers (Kim & Park, 2016). Flight delays may cause a variety of negative emotions in travellers, including frustration, anxiety, and disappointment (Mitsokapas et al., 2021). Delays often cause uncertainty and dissatisfaction among customers (Kim & Park, 2016). Negative feelings such as anger and uncertainty can reduce passengers' intentions to repurchase from the airline and their willingness to positive word-of-mouth communication (Kim & Park, 2016). These emotional responses reduce overall passenger happiness and have consequences for the airline's reputation and consumer loyalty (Mitsokapas et al., 2021). Song, Guo, and Zhuang's (2020) extensive research of SKYTRAX comments provides valuable insights into passengers' emotional responses to flight delays. By analyzing passenger feedback over time, this study demonstrates the diverse range of emotional reactions generated by air travel delays. These emotional responses are crucial as they shape passengers' impressions of the airline and influence their behaviors (Kim & Park, 2016). Kim and Park note that anger, in particular, is a common emotion triggered by service delays, leading to increased dissatisfaction and

negatively impacting customers' overall airline experience. These emotional reactions have a negative effect on passengers' future plans and behaviors (Kim & Park, 2016).

There is a strong and unfavourable relationship between passengers' emotions and their experiences with flight delays, with dissatisfaction being a common response among affected travellers (Song et al, 2020). Thus, airlines and airport authorities need to address the emotional needs of passengers during delays effectively (Kim & Park, 2016). Bogicevic et al. (2017) acknowledged the value of utilizing technology to enhance travellers' experiences at airports, particularly during boarding. Bogicevic et al. (2017) also evaluated how traveller-focused airport technology affected passenger happiness to find ways to improve the entire airport experience.

Addressing the emotional impact of service delays and implementing strategies to meet passengers' emotional needs can help airlines enhance customer happiness, loyalty, and brand reputation, thereby boosting their competitiveness in the aviation industry (Kim & Park, 2016). Neglecting this can lead to negative word-of-mouth, potentially lowering customer loyalty and satisfaction levels in the aviation industry (Song et al., 2020). Understanding passengers' emotional experiences during delays allows airlines to implement strategies to reduce negative emotions, including proactive communication, timely updates, and good customer care (Mitsokapas et al, 2021).

Airlines can significantly minimize disruptions by carefully organizing their operations. Planning tasks such as maintenance, refuelling, and staffing may contribute to fewer delays while enhancing safety and efficiency during layovers (Lan et al., 2006). Reducing airport delays and operationally maintaining longer turnaround and minimum connection times can greatly lower the missing connections. It is possible to prevent missed connections for transfer passengers by keeping the flight turnaround time at 50 minutes, the minimum connection time at 60 minutes, and arrival delays within 70% of the existing spread (Pternea & Haghani, 2019). Optimizing aircraft routing and flight departure schedules contributes to reducing passenger disruptions while improving operational robustness (Lan et al., 2006). This strategic approach ensures that flights are scheduled efficiently, lowering the possibility of delays and improving overall reliability for passengers (Lan et al., 2006).

Missed Flights:

While boarding their connecting flights, passengers face many issues, such as departure or arrival delays, short transfers, and insufficient guidance. Passengers with connections are 2.8 times more likely to be disrupted than local passengers and 1.5 times more likely to be disrupted by a flight cancellation than local passengers (Bratu & Barnhart, 2005). The flight layover system allows for long-distance air travel by connecting flights and giving space to passengers during the time between two flights (Britto et al., 2012). This ensures a smooth movement across flights and assists airlines in handling the movement of travellers, providing

various route options according to traveller requirements (Britto et al., 2012). If a connection is missed at the intermediate airport, passengers must have to wait for the next flight (Cattaneo et al., 2019). The present condition of the layover system encounters certain difficulties, such as bad experiences for passengers, inefficient use of resources, and probable safety issues, which require attention (Roach et al., 2012).

Effects of missed flights on passengers:

Flight disruptions are mainly caused by schedule modifications or related problems that have a significant impact on passenger experiences (Lan et al., 2006). Unplanned layovers cause stress, discomfort, and frustration among passengers (Britto et al., 2012). Delays or cancellations frequently cause passenger anger, resulting in negative word-of-mouth conversations (Jiang et al., 2019). Extended waiting times during layovers can induce stress and uneasiness due to the unpredictability of the situation, impacting passengers' mental health (Roach et al., 2012). Prolonged periods of sitting without movement can lead to discomfort and health issues such as deep vein thrombosis (DVT) due to restricted blood flow (Roach et al., 2012). Layovers, especially if they occur at night or are prolonged, may result in passenger fatigue. This can lead to difficulty sleeping, irritability, and a decline in overall well-being (Roach et al., 2012). Stress from layover experiences can affect the entire journey, leaving passengers feeling depleted upon reaching their final destination. Delays between flights can lead to unavoidable problems like misplaced luggage or discomfort from airport meals, lowering passengers' enthusiasm for future travels in the same settings (Britto et al., 2012). Passengers who missed connections may choose to change airlines or cancel their travel plans, potentially impacting the airline's goodwill. If alternate connections are unavailable on the same day, airlines must accommodate delayed passengers.

Delays associated with layovers can impact passenger satisfaction and have financial implications for both passengers and airlines. Longer layovers could result in higher airfares for flights with layovers, causing a significant challenge (Britto et al., 2012). Customers who prefer to avoid the complications of layovers may be willing to pay more for direct flights, potentially reducing revenue for airlines offering indirect routes. Additionally, airlines may incur additional costs by compensating passengers for extended waiting times or layover delays, which might affect profitability (Britto et al., 2012).

Reasons for connecting flight disruptions:

In transfers, the connection is successful if there is enough time between the departing and arriving flight, and if not, then it fails (Pternea & Haghani, 2019). The delay increases when two subsequent flights share the same aircraft, and departure delays are the primary reason for the delay at the destination airports (Wu et al., 2018). If an aircraft arrives at its scheduled gate but must wait because the preceding aircraft is still occupying that gate, it is referred to as gate blockage. Gate blockage can have many negative impacts, like passenger delays, missed connections, and more fuel burn (Castaing et al., 2016b). In improving passengers' satisfaction,

utilizing gates efficiently plays a crucial role, as a lack of it can lead to serious consequences. (Zhang & Klabjan, 2017). Ali et al. (2019) propose a passenger-centric approach to reduce missed connections at low-cost airports through gate reassignment. The study highlights the importance of optimizing gate allocation to minimize missed connections and aircraft delays. By reallocating gates based on schedule changes and operational risks, airports can enhance operational efficiency and the reliability of air transport (Ali et al., 2019). Implementing adaptive gate management strategies, as suggested, can significantly benefit the aviation sector (Ali et al., 2019).

Bratu and Barnhart's (2005) careful analysis is a sobering account of the circumstances travellers face regarding the differences in the impact of passenger delays on loyal passengers. The paper itself corresponds to the fact that the flight schedule is ineffective. They examined delays in a major US hub-and-spoke airline. Their analysis of passenger delay data gathered from flight operations and booking datasets revealed significant differences in flight leg delays, missed connections, and passenger outcomes. Using Passenger Disruption Control (PDC), they strategically rescheduled passengers affected by delays or cancellations to minimize missed connections and prioritize recovery efforts and passenger satisfaction. Their study underscores the importance of aligning scheduling methods with passenger behaviors and needs, offering insights to enhance recovery efforts. It is a clear indication that there is no other alternative but to move to overhaul the timetable and traffic management systems to make life bearable for passengers.

Potential ways to minimize layovers and missed flights:

Lan et al. (2006) highlight the importance of proactive scheduling methods to prevent passenger inconvenience caused by disruptions. The researchers emphasize the importance of integrating flexibility into airline schedules by improving aircraft routings and flight departure times (Lan et al., 2006). This method involves anticipating potential difficulties and planning routes and schedules to handle delays without severely affecting connecting flights. They suggest a strategy for optimizing aircraft routing that takes into account the risk of delays, intending to minimize the overall expected delay spread over the network during disruptions (Lan et al., 2006). Furthermore, the study investigates the concept of scheduling aircraft legs in shorter periods to prepare for unexpected delays (Lan et al., 2006).

On the other hand, Cattaneo et al. (2019) focus on the weakness of open traffic in the European air transport system, which allows for complexities and weak points in connecting flights that usually take place without the use of a traditional hub-spoke system. By setting these times on the rise of budget carriers, analysts should be aware of whether these connections threaten the continuity and sustainability of air travel.

Moreover, deviating from the study of Ali et al. (2019), the authors show that stochastic delays can affect the dimension of missed connections at airports, especially at low-cost ones. In this

case, it is necessary to actively approach the mitigation of failures and the development of operational resilience. The data show how the unexpected relationships between schedule elements and the needs of airport participants demonstrate the need to take the necessary steps to achieve these unpleasant effects.

As a result, Pternea and Haghani (2019) bring together key aspects that include the effects of aircraft-to-aircraft gate movements as an influencing factor in mitigating the downstream effects of delays. With their framework, airport authorities have a strategic tool at their disposal to increase operational resilience and passenger fulfillment when passenger schedules are disrupted due to unforeseen causes.

Airlines and airports can enhance on-time performance and minimize disruptions by utilizing advanced weather forecasting systems and data-driven scheduling algorithms (Borsky & Unterberger, 2019). Furthermore, investing in infrastructure improvements and operational flexibility can increase resistance to unfavorable weather conditions, lowering the social and economic costs of flight delays. Through strategic collaboration and innovative ideas, the aviation industry may establish a more resilient and efficient system that can resist the challenges posed by unpredictable weather events (Borsky & Unterberger, 2019). Mitsokapas et al. (2021) address alternative solutions to flight delays, emphasizing the need to tackle this major problem in the aviation industry. By evaluating airline delays, the study emphasizes the importance of preventive measures to reduce disruptions and improve operational efficiency (Mitsokapas et al., 2021). Implementing advanced weather forecasting systems and data-driven scheduling algorithms can assist airlines in anticipating and responding to unfavorable weather conditions, hence decreasing severe weather delays. Furthermore, investment in infrastructure upgrades and operational flexibility can boost resilience to disruptions, lowering the social and economic costs of flight delays (Mitsokapas et al., 2021). Through strategic collaboration and innovative ideas, stakeholders in the aviation industry can work together to create a more resilient and efficient system that can survive the challenges provided by unexpected occurrences while also improving overall customer satisfaction (Mitsokapas et al., 2021).

Together, these studies offer a detailed overview of the upheavals facing the aviation industry, while reinforcing the need for new solutions as a means of limiting disruption to air travel. By examining the various elements that contribute to these delays and cancellations, this review will reveal ways in which policymakers and stakeholders can work together to strengthen a robust passenger-centric transport future.

Advanced Communication Strategies:

Effective communication between airlines, airports, and passengers can greatly enhance the layover experience. For instance, providing real-time updates on flight changes, delays, or boarding processes can minimize passengers' stress levels and increase overall satisfaction

(Jiang & Barnhart, 2009). Furthermore, proactive communication enables passengers to more effectively plan their layovers and make well-informed travel decisions.

Policies

The Canadian Transportation Agency (CTA) created policies to protect the rights of air travel passengers (Toolkit, 2023). These policies cover different situations that might occur when travelling by air, such as compensation when a flight disruption happens, the standard of treatment for passengers during a flight disruption, denied boarding, lost or damaged baggage, tarmac delays, etc. The protection of passengers' rights and interests in the event of flight delays is not only related to the interests of passengers themselves but also hinders the overall development of the civil aviation industry (Geng, 2023). The judgment standard of flight delay or missed connection is a complex area because passengers and airlines have very different understandings of what a "flight delay" means or the reasons behind it. Airlines are required to keep passengers regularly informed if there is a flight disruption; they tell passengers why their flight has been disrupted (Toolkit, 2023). Although the airline informs passengers, little information is given about the reason for the flight disruption. This might be done to prevent passengers from getting anxious or worried, but the lack of information causes the passengers to get angry or dissatisfied. It is understandable for delays to occur because air carriers can not always arrive at their destinations at the exact time of their flight schedule (Geng, 2023).

However, when the delay becomes 3 hours or more, the CTA expects the airline to compensate passengers, which may put pressure on the airlines. But when passengers do not know about their rights, they may lose the opportunity to receive compensation, and airlines sometimes do not inform passengers that they are entitled to compensation.

Although the Canadian Transport Agency (CTA) has created the Air Passenger Protection Regulation and released it to the public, many people remain unaware of their rights (The Canadian Press, 2019). This shows that there is indeed a communication gap that needs to be addressed. It is difficult to determine where the communication gap occurs between the CTA, the airlines, and the passengers. The CTA has the responsibility to create and enforce the policies or regulations for the airlines and passengers, the airline has the responsibility to inform passengers about their rights, and the passengers have the responsibility to learn and understand their rights. But, as long as a communication gap continues to exist in the different roles, it will be difficult to improve passenger's experiences.

Compensating passengers for delays or missed flights is only one section of the air passenger's protection rights, which is why this study will focus on the following categories: Passenger rights, which state that when an airline delays or cancels a flight, what passengers are entitled to depends on the level of control that the airline has over the reason for the delay or cancellation. Communication, which states that the airline has to give information about why the flight was delayed or cancelled, what assistance they have to give passengers, if any, the compensation

passengers may be entitled to for your inconvenience, and what recourse is available to passengers: compensation, Compensation, Refund or rebooking, and assistance (Toolkit, 2023). This study aims to determine and understand the reason for the communication. gap between the service provided and to understand if the appropriate level of service is received by passengers when a flight disruption happens.

Section 2: Research Methods and Research Results

Part I: Research Question and Methods

The research question

What are the factors contributing to the communication gap between the services provided to passengers and their expectations during missed connecting flights, particularly concerning passenger rights and compensation?

Our research question is aimed at identifying the reasons behind the communication gap between airline services and passenger expectations during missed connecting flights, with a focus on understanding why passengers often remain uninformed about their rights and compensation, aiming to improve communication, reduce frustration, and enhance overall customer satisfaction and trust in airlines.

Objectives (Sub-questions)

- Identifying information barriers for passengers and how clarity impacts uncertainty during flight disruptions.
 - We aim to understand how clear communication can minimize passenger uncertainty and improve their experience during flight disruptions.
- Analyze how previous experiences and industry standards influence passenger expectations.
 - We want to understand how passengers' past experiences and industry benchmarks shape their expectations, which can help airlines tailor their services to meet or exceed those expectations.
- Examining the airline policies and passenger rights to enhance cooperation.
 - We aim to identify gaps and areas for improvement, ensuring that passengers feel supported and informed, which can lead to smoother operations and increased trust in the airline.
- Assessing whether passengers' experience differs from what they expect during a missed connection.

- We want to identify gaps between passengers' expectations and their experiences during a missed connection. This can help airlines address shortcomings and improve service quality to better meet passenger needs.

Answering these questions will provide valuable insights into how airlines can enhance passenger experiences during missed connections. It will also inform efforts to effectively communicate passenger rights and improve compensation processes, ensuring fairness and transparency in handling travel disruptions. Addressing these areas has the potential to strengthen customer relationships and build trust in airline services.

Part II: Research Results and Recommendations

Questionnaire Findings:

The online questionnaire was active for over three weeks, and we received over 100 responses. After cleaning the data, 107 responses were considered valid, with 66 of them matching our target demographic. (see “Limitations of the Results”).

Demographics and Situational Characteristics:

Age of Participant	No of responses	% of responses
18 - 30	97	90.7%
31 - 40	09	8.4%
41 - 50	0	0.0%
51 and more years	01	0.9%

Table 1: Participant responses- age

Travelled alone in past 2 years	No of response	% of response
Yes	90	84.1%
No	16	15.0%
I prefer not to say	01	0.9%

Table 2: Participant responses- solo travelers

Travelled internationally in economy class	No of response	% of response
Yes	88	82.2%
No	18	16.8%
I prefer not to say	01	0.9%

Table 3: Participant responses- internationally in economy class

Number of connecting flight	No. of responses	% of responses
1	40	37.4%
2	35	32.7%
3 or more	16	15.0%
No connecting flight	15	14.0%
I prefer not to say	1	0.9%

Table 3: Participant responses- no. of connecting flights

Experienced disrupted connecting flight	No. of responses	% of responses
Yes	67	62.6%
No	39	36.4%
I prefer not to say	01	0.9%

Table 4: Participant responses- experience of disrupted flights

The findings from the tables show a strong focus on younger participants in our study on air travel experiences. Specifically, most participants (90.7%) were between 18 and 30 years old. This suggests that our survey mainly captured the perspectives of younger adults who have traveled internationally alone in the past two years. In comparison, participation dropped sharply in older age groups: those aged 31 to 40 made up 8.4% of respondents, while those aged 41 to 50 and 51 years and older accounted for very small percentages (0% and 0.9%, respectively). This distribution shows a potential limitation in the age diversity of our study, with middle-aged and older individuals being underrepresented.

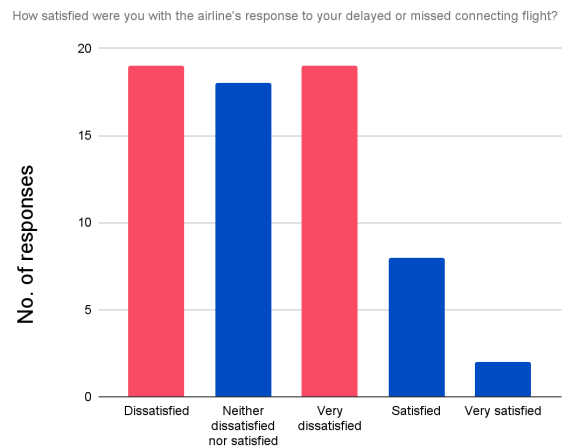
This also shows varying levels of complexity in participants' travel itineraries, with many encountering multiple connecting flights. Most participants had one or two connecting flights,

indicating relatively straightforward travel routes. However, some respondents had three or more connecting flights, highlighting a group of travelers with more complex and potentially challenging travel schedules. Understanding these patterns is important for assessing the overall travel experience and identifying possible stress points linked to different levels of connectivity. This also shows the diversity in travel patterns among respondents, which may affect how airline services and airport facilities are perceived and used in different travel situations.

While this research mainly reflects the experiences and views of younger travelers, future studies could benefit from strategies to include a more balanced range of ages. This would help provide a more complete understanding of air travel experiences across different age groups.

General Findings

We have included charts to show the general findings from the survey below:

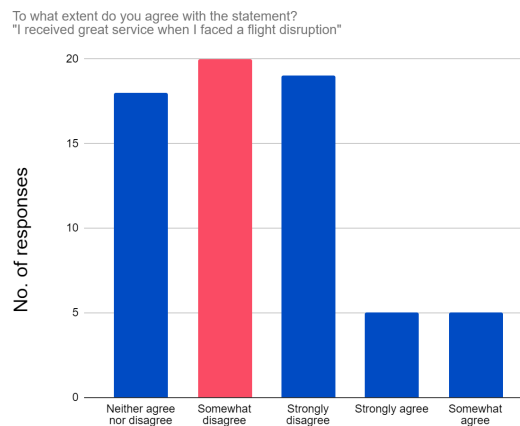


Img.1: Count of satisfaction with airline’s responses

The survey results reveal a significant level of dissatisfaction among passengers with how airlines handled flight disruptions. Specifically, 28.8% of respondents reported being very dissatisfied, and 27.3% were dissatisfied, resulting in a combined total of 56.1% of passengers who were unhappy with the airlines' responses. This indicates that more than half of the passengers did not find the airlines' actions adequate or satisfactory in response to flight disruptions.

Conversely, 28.8% of respondents were neutral, suggesting that while they did not have strong negative feelings about the response, they also did not find it particularly positive. This neutrality may reflect indifference or a sense of resigned acceptance among these passengers.

A smaller portion of passengers expressed satisfaction with how airlines manage disruptions. Only 3% of respondents were very satisfied, and 12.1% were satisfied, making up a total of 15.1% who had a positive experience. This minority suggests that only a small fraction of passengers felt that the airlines met or exceeded their expectations in handling flight disruptions.



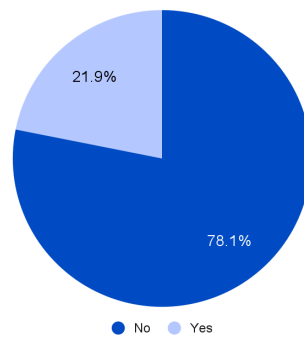
Img.2: Graph for the responses of 'Do you agree that you received great service?'

The survey results provide insights into passengers' opinions on the statement, "I received great service when I faced a flight disruption," revealing a clear trend of disagreement among the majority. Specifically, 28.4% of respondents strongly disagree with the statement, and 31.3% somewhat disagree, resulting in a combined 59.7% who do not agree. This majority indicates that more than half of the passengers hold a negative view toward the statement, reflecting a strong consensus against it.

Additionally, 23.9% of respondents neither agree nor disagree with the statement, indicating a significant portion of passengers who are ambivalent or undecided. This neutrality might suggest a lack of strong feelings or insufficient information to form a clear opinion on the matter.

On the other hand, 7.5% of respondents somewhat agree, and 9% strongly agree with the statement, totaling 16.5% who are in favor. This minority indicates that a smaller fraction of passengers find the statement to be accurate or acceptable. Despite being a smaller group, their agreement shows that some passengers still hold a positive view of the service received during flight disruptions.

Have you read the Air Passenger Protection Regulations?



Img.3: Graph of responses to 'Have you read Air Passenger Protection Regulations?'

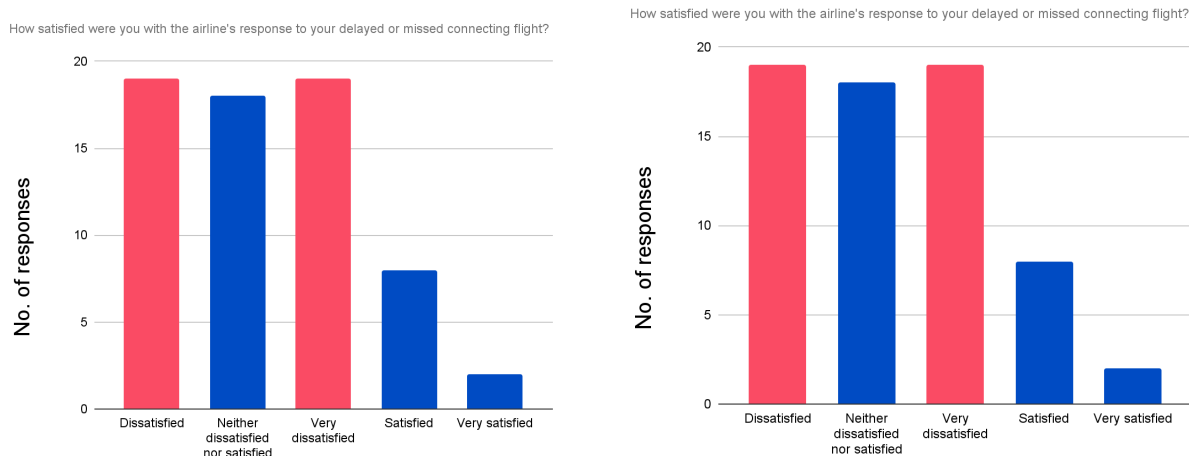
The survey results highlight a significant lack of awareness among passengers regarding the Air Passenger Protection Regulation. An overwhelming 76.1% of respondents admitted they have not read the regulation, indicating that most passengers are unfamiliar with their rights and protections related to air travel.

In contrast, only 19.4% of respondents reported having read the Air Passenger Protection Regulation. This minority suggests that fewer than one in five passengers have taken the time to understand the regulatory framework intended to protect their rights. This lack of knowledge may result in passengers not seeking compensation or failing to advocate effectively for themselves during travel disruptions. It also implies that airlines or relevant authorities are not successfully communicating this critical information to passengers.

The 4.5% of respondents who remained neutral likely represent individuals who are either unsure if they have read the regulation or have some limited awareness without fully understanding its details.

Analyzing Relationships:

Looking at the findings, we analyzed the relationship between the findings and the results.



Img.4: The relationship between graphs

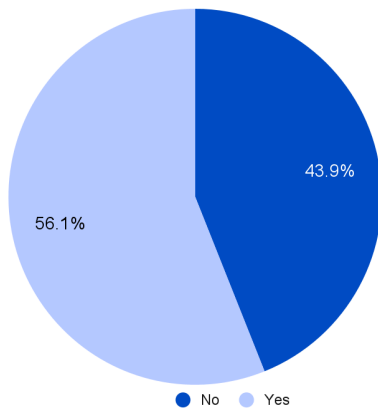
The relationship between the two charts provides valuable insights into passenger perceptions and experiences.

Firstly, the chart showing satisfaction with the airline's response reveals that a majority of participants were dissatisfied. Specifically, 56.1% expressed some level of dissatisfaction, with 28.4% strongly disagreeing and 27.3% somewhat disagreeing with the airline's handling of the situation. This reflects a prevalent sentiment among passengers that airlines did not adequately meet their expectations or needs during flight disruptions.

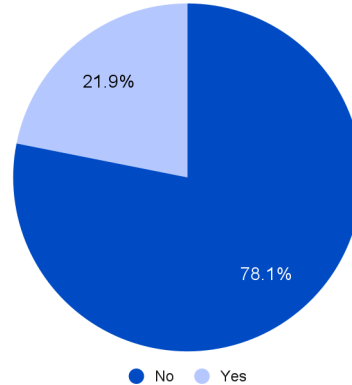
Secondly, the chart measuring agreement with the statement "I received great service when I faced a flight disruption" shows a similar trend. A significant majority, 59.7%, either strongly disagree (28.4%) or somewhat disagree (31.3%) with the statement. This indicates that most participants did not feel they received excellent service from the airlines during flight disruptions.

The relationship between these two sets of data suggests a strong correlation: passengers who were dissatisfied with the airline's response to flight disruptions also tended to disagree with the statement about receiving great service. This connection implies that there is a significant overlap between passengers who felt let down by the airlines' handling of disruptions and those who did not perceive the service they received during such disruptions as satisfactory.

Knowledge of entitlement to compensation?



Have you read the Air Passenger Protection Regulations?



Img.5: The relationship between pie charts

The survey results from the two charts provide insight into the relationship between passengers' knowledge of their entitlement to compensation and their awareness of the Air Passenger Protection Regulation. Understanding this relationship is crucial for grasping the broader context of passenger rights and awareness.

The first chart shows that 53.7% of respondents are aware of their entitlement to compensation under certain conditions. This suggests that over half of the passengers have some understanding of their rights to compensation when flight disruptions occur. This awareness could be attributed to personal experiences, word-of-mouth, or information provided by airlines or other sources.

However, the second chart reveals a different scenario when it comes to detailed knowledge of the Air Passenger Protection Regulation itself. Only 19.4% of respondents have actually read the regulation, indicating that a much smaller percentage of passengers are familiar with the specific legal framework and the detailed protections it offers. This highlights a significant gap between general awareness of compensation entitlements and a deeper understanding of the regulatory details.

Moreover, 44.8% of respondents do not know when they are entitled to compensation, a figure that aligns closely with the 76.1% who have not read the Air Passenger Protection Regulation. This suggests that a lack of detailed knowledge about the regulation correlates with a lack of awareness about specific entitlements. Without reading the regulation, passengers are less likely to understand the exact conditions under which they can claim compensation, leading to confusion or missed opportunities.

The relationship between these two sets of data underscores a critical gap in passenger education and awareness. While a moderate majority (53.7%) know about their entitlement to

compensation in a general sense, the lack of detailed knowledge (with only 19.4% having read the regulation) suggests that passengers might not fully understand the scope of their rights or the specific processes involved in claiming compensation. Furthermore, the 44.8% who do not know about their entitlement to compensation likely belong to the larger group that has not read the regulation. This indicates a clear need for improved passenger education and awareness efforts.

Interview Findings

What we heard: Participant Interviews

After transcribing and analyzing data from participant interviews, we discovered the following common themes from participants;

Theme 1: Unclear communication

Participants report that they do not receive clear explanations or updates regarding the reasons for flight delays from airlines. This lack of information leads to feelings of being uninformed and frustrated. Without understanding the cause of delays, passengers may feel powerless and uncertain about how long they will be delayed or what actions, if any, they can take to mitigate the impact on their travel plans.

Additionally, language barriers can further complicate the situation, especially when passengers are connecting through airports in different countries with diverse languages. These barriers can make it difficult for airline staff to communicate effectively with passengers, leading to misunderstandings or incomplete comprehension of the situation. This can increase feelings of frustration and confusion among passengers who are already stressed by travel disruptions.

Participants also express a strong preference for airlines to provide clear and proactive information during flight disruptions. This includes timely updates about delays, explanations for the disruptions, and realistic estimates of when flights are expected to resume or depart. Proactive communication would help passengers feel more in control and informed, potentially reducing anxiety and improving their overall perception of the airline's customer service.

"They can be honest to us, like, yeah, if the flight is going to be delayed, so just straight away say that on our face."

-Participant A

"The layover was in Paris, so most of the staff usually speak a different language, so there was no point of contact, like physically, and there was no assurance like how we can just talk to them."

-Participant B

Theme 2: Compensation issues

Many passengers are not fully aware of their rights concerning compensation for flight delays or cancellations. This includes understanding eligibility criteria, such as minimum delay durations and specific circumstances under which compensation applies. This lack of awareness can result in passengers either not claiming the compensation they are entitled to or misunderstanding their entitlements, leading to frustration and dissatisfaction with airline service.

A common misconception among passengers is that airlines are always liable for flight delays or disruptions. Some travelers may misinterpret consumer protection laws or airline policies, assuming they are entitled to compensation under all circumstances. This misunderstanding can lead to unrealistic expectations and disappointment when airlines do not provide compensation that passengers mistakenly believe they are owed.

During flight disruptions, such as significant delays or cancellations, passengers sometimes find themselves needing to arrange accommodation independently. This situation often arises due to a lack of immediate assistance or clear communication from the airline about their responsibilities for passenger welfare during such events. The absence of timely support from the airline forces passengers to incur additional expenses and inconvenience, further increasing their dissatisfaction.

"So it's like what needs to happen for me to have the right to get the compensation."

-Participant C

"At least I feel that in my case, the company was liable for what happened."

-Participant D

Theme 3: Knowledge of air passenger protection rights

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"We don't think that we're going to need that anytime soon, so we're just like, okay, yeah, I accept everything."

-Participant E

"...It's like so much to read through, it's like reading a textbook It would be better if important points were highlighted."

-Participant F

Theme 4: Reasons for delays

Flight delays can result from various factors such as weather conditions, technical issues, crew scheduling problems, or air traffic control issues. These reasons may change as airlines receive updated information or face new challenges. Passengers often become frustrated when they perceive a lack of consistent and clear communication regarding the specific reasons for delays. This inconsistency can lead to confusion and a sense of distrust towards the airline's communication practices.

Some passengers believe that airlines are not always transparent or honest about the reasons for flight delays. This perception can arise when passengers receive vague or conflicting explanations or when they suspect that airlines might withhold information to avoid liability or negative perceptions. The absence of clear and timely communication about delay causes can increase feelings of mistrust among passengers.

Passengers express a strong preference for airlines to provide detailed emails or notifications explaining the specific reasons for a flight delay and its expected duration. Clear and comprehensive communication helps passengers better understand the situation, manage their expectations, and make informed decisions about alternative arrangements or accommodations. Providing detailed explanations also shows transparency and accountability from the airline, which can enhance passenger satisfaction and trust.

"knowing nothing is more concerning than knowing the truth."

-Participant G

"We felt a bit tricked because they were trying to hide the truth about what was going on."

-Participant H

Theme 5: Future travel expectations

When passengers experience delays, cancellations, poor customer service, or other negative incidents, it often results in a loss of trust in the airline. They may perceive the airline as unreliable or unresponsive to their needs. This loss of trust can lead passengers to actively avoid booking with that airline in the future, preferring competitors that they perceive as more reliable or accommodating. We also discovered that this negative experience can lead to passengers comparing their experiences across different airlines. Negative encounters with one airline may prompt passengers to evaluate alternatives more critically. If they perceive that another airline consistently offers better service, reliability, or customer support, they may choose to switch their loyalty to that competitor for future travel plans. We discovered that negative experiences leave a lasting impression on passengers' travel decisions, and they usually continue to remember that experience. They become more cautious when booking flights, considering factors beyond price or convenience. Factors such as customer reviews, reputation for reliability, and previous personal experiences with the airline weigh heavily in their decision-making process. A single negative experience can influence their perceptions and behaviors for an extended period, impacting their loyalty and willingness to choose that airline again.

"it influences a lot... I'm not now reluctant to take a plane in that sense, but maybe I will think twice if it is going to be with this company specifically."

-Participant J

What we heard: SME Interviews

We interviewed 4 SMEs for this study; two were professors in aircraft maintenance, one was a pilot, and the other worked as a customer service/operations manager. They gave us a lot of important information that helped to validate our findings.

Academic SMEs:

They both had expertise in aircraft maintenance, offering insights from both a participant's and a subject matter expert's (SME) perspective. We consulted them to understand the underlying causes of flight disruptions and the behind-the-scenes factors involved. Typically, bad weather and maintenance issues are cited as the most common reasons for flight disruptions. However, the SMEs highlighted that maintenance may not always be the true cause; airlines might use it as a blanket reason to avoid providing detailed explanations. In some cases, flight disruptions

occur because the pilot has reached their maximum allowable hours and no replacement crew is available, or due to delays in loading baggage onto the aircraft.

Flight disruptions often result from a combination of factors, which can cascade into multiple smaller issues, ultimately leading to a negative experience for passengers. The SMEs also pointed out that there are restrictions on how much information airline staff can share with passengers during a disruption. Providing too many details could make passengers anxious or fearful, and giving an exact disruption time could lead passengers to leave the gate area, creating challenges if the aircraft becomes ready earlier than expected.

Regarding air passenger protection rights, the SMEs mentioned that while airlines include these in their terms and conditions, they do not actively inform passengers about their rights, leading to a lack of awareness.

Industry SMEs:

The pilot and customer service SMEs provided extensive insights from an industry perspective. They both highlighted a communication gap between what is conveyed to passengers and what passengers understand. Airlines are working to reduce flight disruptions, as seen through collaborations between large and smaller domestic airlines, but there is still much progress to be made. The SMEs emphasized that experience and knowledge are crucial when dealing with passengers during a flight disruption.

The interview with the SMEs also revealed that airlines do not actively inform passengers about their rights. Consequently, passengers are often unaware of the criteria that make them eligible for compensation during a flight disruption. This communication gap is a significant issue, as passengers do not understand the existing limitations. Additionally, it is important to remember that every airline is different, and passenger experiences can vary from negative to positive depending on the airline.

The SMEs suggested that making air passenger protection rights more accessible and discoverable for passengers would help address these issues.

Key Insights

After combining findings from quantitative data (i.e., surveys) and qualitative data (i.e., participant and SME interviews), we discovered these key insights:

Insight 1

Passengers tend to lose trust in airlines due to insufficient communication, as airlines often face limitations in disclosing details about flight disruptions.

According to interview findings, participants expressed losing trust in airlines and feeling deceived at times. We found out that participants do not have an issue with the flight disruption,

but what happens during the flight disruption, particularly communication? An SME highlighted that airlines face constraints on the amount of information they can disclose to passengers, especially concerning the duration of delays. For instance, if passengers are informed of a 7-hour delay, some may decide to go do something else outside the airport. However, there are instances where such delays may be resolved sooner than anticipated, posing logistical challenges in locating passengers promptly for boarding.

Insight 2

Passengers are unaware of their rights because there are no proactive measures to educate them about passenger rights.

Both secondary research, quantitative data, and qualitative data indicate a widespread lack of awareness among passengers regarding air passenger protection regulations. Many participants did not know about the existence of this document, even though the airlines mention it in their terms and conditions document (some airlines also mention it on the ticket). However, there are no proactive measures in place to inform passengers specifically about their rights and protections. Consequently, passengers often do not realize the benefits available to them when faced with flight disruptions. An SME noted from personal experience that airlines rarely, if ever, mention passenger rights during travel, highlighting a significant gap in communication and education on this issue.

Insight 3

Passengers find the rights and regulations document difficult to understand because it has a lot of information.

Participants expressed difficulty in understanding the document, with one likening it to reading a textbook. An SME confirmed that locating and comprehending the document is challenging. This may possibly be due to complex wording and the overwhelming amount of information provided. This contributes to passengers finding it difficult to navigate and access the specific information they need.

Insight 4

Passengers are unsure about their eligibility for compensation because it is difficult to understand and find this information.

Passengers sometimes believe that they are entitled to compensation when a flight disruption happens, but apparently, there are certain criteria for them to be entitled to compensation (i.e., disruptions within the airline's control). This lack of knowledge leads to the misconception that every disruption qualifies for compensation. Furthermore, even when passengers are aware of their entitlement, many are uncertain about how to actually claim compensation. In some cases, passengers rely on others who are familiar with the process to guide them. Additionally, passengers struggle to locate information about compensation within air passenger protection regulations, further complicating their ability to understand and assert their rights.

Limitations of result

Although the survey included verification questions, we identified some insincere responses, likely due to participants completing the survey multiple times and providing random answers in hopes of winning a gift card. We chose not to restrict responses to one per participant to avoid collecting email addresses, which would have violated our confidentiality guarantee as outlined in the informed consent form. We took steps to clean the data and removed suspicious entries, but we cannot ensure the dataset is entirely accurate. Therefore, the results should be interpreted with caution.

Additionally, we were unable to conduct an interview with a subject matter expert (SME) from the help desk, who would have had direct contact with passengers. This omission means we missed out on valuable firsthand knowledge from a potentially critical group for our intervention. Furthermore, we only interviewed four SMEs, which may not provide a comprehensive perspective on the research findings. However, we were able to validate our findings with participants and received a common statement from the four SMEs, which helped guide our key insights.

Despite these limitations, we believe our study offers relevant insights. It contributes to a case study that can be used to enhance passenger experiences during flight disruptions and increase awareness of Air Passenger Protection Rights.

Experience Principles

From the perspective of the passenger, what would they want to be done to improve their travel experience, especially during flight disruptions? We have highlighted these in the form of experience principles that will influence our future ideas.

Principle 1: I want transparency about flight disruptions.

Persona's reasoning

Transparency about flight disruptions is crucial to me because it allows me to make informed decisions about my travel plans. When airlines communicate openly and honestly about the reasons for delays or cancellations, it helps me understand the situation better. Knowing the expected duration of a delay and the steps the airline is taking to resolve the issue gives me confidence that my time and travel experience are respected. Transparent communication also builds trust between me as a passenger and the airline, showing their commitment to keeping me informed and minimizing the inconvenience caused by disruptions.

So what?

I want airlines to clearly communicate the exact reason for any disruptions so that I can continue to trust them.

How might we as researchers:

Improve airline communication to ensure passengers are well-informed about flight disruptions, despite the limitations in disclosing details?

Principle 2: I want to know my rights and where I can find them.

Persona's reasoning

As a passenger, knowing my rights and where to find them is essential for navigating any potential issues or disruptions during air travel. Knowing my rights means being informed about what protections and entitlements I have under airline policies and consumer protection regulations. Being aware of the legal framework governing air passenger rights helps me advocate for myself if issues arise.

So what?

I want to know my rights as a passenger and easily find information on them, so I can be confident if a flight disruption occurs, knowing that I am equipped to address any challenges that may arise and advocate for fair treatment and compensation when necessary.

How might we as researchers

Create awareness about air passenger protection rights?

Principle 3: I want it to be simple for me to understand.

Persona's reasoning

As a traveler, simplicity in understanding my rights is crucial for a smooth and informed travel experience. I value straightforward and easy-to-access information about what I'm entitled to as an air passenger.

So what?

I want the rights and regulations document simplified so that I can easily understand the information, thereby reducing uncertainty.

How might we as researchers

Make the rights and regulations document more understandable for passengers, despite containing a lot of information?

Principle 4: I want to know if I am eligible for compensation and how to receive it.

Persona's reasoning

As a passenger, I would appreciate having clear and understandable criteria that outline when I am entitled to compensation, such as for flight delays, cancellations, or denied boarding. Knowing these criteria upfront helps me assess my situation accurately. It is also essential for me to easily access information about the compensation process. This includes understanding what documentation or evidence I may need to provide, how to submit a claim, and any deadlines or procedures involved.

So what?

I want to know when I am eligible for compensation, what steps I need to take to obtain it from the airlines for flight disruptions, and how to exercise my rights.

How might we as researchers

Simplify the information regarding compensation eligibility, and make it more accessible for passengers?

Section 3: Solution Development, Testing, and Implementation

Personas

We created two personas out of the demographics that we had for this project. Let's now take a look at what those personas look like:

1. Priya



Priya is a 25-year-old student from India. She is coming to Canada to begin her higher education. She is excited and enthusiastic about this new experience. She is anxious and curious, mainly because she is going to a new country on her own. She is technologically savvy; because of her age, she does not find it difficult using modern technology. Her goals are to arrive in Canada without any delays, have a smooth and comfortable travel experience, and get to Canada early because her student orientation is in two days. Her current painpoints are travel-related anxiety and excitement due to this being her first international trip, worry about missing flights and additional security procedures, and fear of travelling alone all by herself.

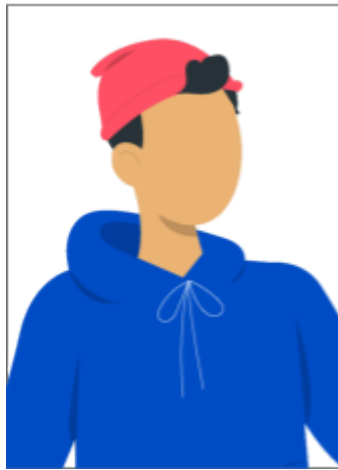
Unfortunately, the journey was not as smooth as she would have hoped for because one of her flights was significantly delayed, and the airline provided poor communication and minimal support. She did not receive any accommodation, meals, or coupons for the delay.

Priya's persona helped shape the solution by highlighting the key pain points that many travelers face during disrupted flights, especially those who are unfamiliar with their rights. Her experience of poor communication and lack of awareness of passenger rights underscores the need for a way to create awareness and a user-friendly platform that simplifies complex

regulations like APPR. Her anxiety and concern about missing connecting flights and dealing with additional security procedures emphasize the importance of having accessible, clear, and supportive resources available during such stressful situations.

Understanding Priya's goals, such as ensuring a timely arrival for her higher education and having a smooth travel experience, drives the need for a solution that not only provides information but also offers practical support. The focus on creating a website with a simplified APPR, along with awareness materials in airports, directly addresses the gaps she faced. This solution aims to reduce the anxiety and uncertainty she and other travelers may experience by equipping them with the knowledge and tools needed to handle flight disruptions effectively.

2. Jose



Jose is a 24-year-old job seeker from Spain who now lives in Canada. He is travelling to Spain for a few days to attend his brother's wedding. He is patient and curious and is in continuous communication with his family about the trip. His goals are to buy an affordable flight ticket to Spain, book the flight with a well-known airline to have a good flight experience because he believes that the more well-known an airline is, then the better the flight experience, and he wants to spend quality time with the family and friends back in Spain. His current painpoints are excitement tempered by concerns about potential travel issues and fear of missing his flight and facing delays during his travel. He was very excited for his journey. Unfortunately, his first flight got delayed. When he reached the connecting airport, he was informed that his connecting flight

had left without him. He was worried. From his past travels, he knew that he could ask for compensation in such a situation. Beaming with hope, he asks for compensation from airlines. The airline informs him that he is not eligible for compensation. Jose got irritated with the response and found the rights confusing.

Jose's persona helped in the ideation of the solution by highlighting the complexity and confusion surrounding passenger rights and compensation eligibility. His experience of being denied compensation despite his understanding of the rules revealed the need for a clear, accessible resource that demystifies these rights for travelers. His frustration and irritation underscore the importance of providing passengers with straightforward information and tools, like a compensation eligibility checker, to quickly assess their situation.

His goals, such as securing affordable flights while maintaining a good travel experience, emphasize the need for a solution that not only clarifies passenger rights but also enhances the overall travel experience by reducing stress and uncertainty. The solution, which includes a user-friendly website and compensation eligibility checker, aims to empower travelers like Jose with the knowledge they need to navigate disruptions confidently and effectively, ensuring that

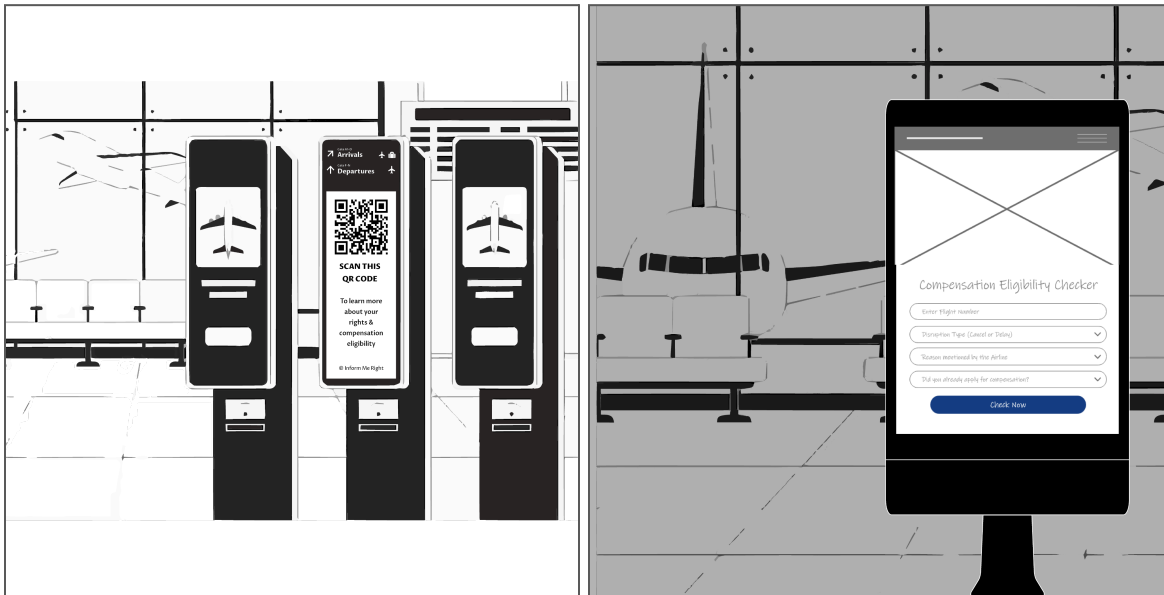
their personal goals, like spending quality time with family, are not compromised by unexpected travel issues.

Journey Map

We created a journey map, which is an analysis of the user's actions, behaviors, touchpoints, painpoints, and opportunities throughout the different phases of the journey. The following are the painpoints and opportunities on the Journey Map by the phases:

- Phase 1: Passengers do not read the terms and conditions provided by the airlines, which mention their rights. Many do not know the document informs them of their rights. Opportunity: **Create awareness** that passengers rights are also included in the terms and conditions document that the airlines give to passengers.
- Phase 2: There is little to no proactive measure being used in the airport to inform passengers about their rights. Opportunity: **Create awareness** by posting posters in the airports with a barcode that would lead to the air passenger rights website.
- Phase 3: **Create awareness** by posting posters in the airports with a barcode that would lead to the air passenger rights website.
- Phase 4: **Create awareness** by having ads on the in-flight entertainment during the flight.
- Phase 5: Passengers want the truth about why their flight is disrupted, and when a delay continues to increase, with no information, they start to assume that the airline is lying to them. Opportunity: **Improve communication** and inform passengers about flight disruptions and the reason.
- Phase 6: Passengers get information from other passengers that may not be accurate, which would create an expectation in the minds of other passengers. Opportunity: **Create awareness** by having ads on the in-flight entertainment during the flight.
- Phase 7: Passengers do not understand when they should be eligible for compensation and some passengers struggle to find and understand the APPR hard to understand. Opportunities: Make the APPR **simple** to find and understand for passengers. Help passengers **understand** when they are eligible for compensation.

places at the airport like on the screens, behind the seats, and on brochures at the help desks.



- When scanned, the QR Code takes passengers to a webpage where they see two options. The first option is to **Read Full APPR** and another option is **Compensation Eligibility Checker**.

[Home](#) [About](#) [Eligibility Check](#) [APPR](#)

Inform Me Right

Helping **Passengers** know their **rights!**

Did you face a disruption and are confused what steps to take next? Or are you curious what rights do you have as a air traveler? We are here to help you!

[Compensation Eligibility Checker](#)

[Read Simplified APPR](#)



SCROLL DOWN

- If the user selects **Read Full APPR**, they will be redirected to a section on the website where they will be able to read the simplified version of APPR and learn more about the rights that they have.

	Home	Eligibility Checker	Make a complaint	APPR
--	------	---------------------	------------------	------

Home > My flight was delayed

My flight was delayed or cancelled

Your Rights

When an airline delays or cancels a flight, what you are entitled to depends on the level of control that the airline has over the reason for the delay or cancellation. The situation can either be:

- ▶ Within the airline's control
- ▶ Within the airline's control but required for safety
- ▶ Outside the airline's control

During a flight disruption

Airlines will be required to keep passengers regularly informed if there is a flight disruption (flight and tarmac delays; flight cancellations; and denials of boarding). They will have to tell passengers why their flight has been disrupted as soon as is feasible.

This information will have to be provided through:

- an audible announcement;
- a visible announcement, upon request; and
- the available communication method the passenger has selected (e.g., email, SMS).

Airlines will have to provide flight status updates every 30 minutes until a new departure time has been confirmed. The airline must offer any new status information to passengers as soon as is feasible, which may be sooner than 30 minutes after the last update.

Compensation

This ONLY applies for delays and cancellations fully within the airline's control.

The airline must pay you compensation for your inconvenience if:

- the delay or cancellation is fully within the airline's control
- you were informed of the delay or cancellation 14 days or less before your original departure time
- you arrived late at your final destination by 3 hours or more, and
- you filed a request for compensation in writing with the airline within 1 year of the delay or cancellation

Compensation Level

The amount of compensation owed depends on how late you arrived at your final destination and on the size of the airline.

- We are not going to reword anything in the APPR document, but what we are going to do is add explanations and some graphics to the content for better clarity to make it easier for the passenger to understand.

Understand your Rights as a Passenger

The Air Passenger Protection Regulation (APPR) applies to **all** flights to, from, and within Canada. This includes connecting flights. If you experienced a problem, you must first contact your airline to resolve it. If that does not work, you can file a complaint with us. The CTA processes complaints and settles disputes between travelers and airlines.

[Read the Full APPR](#)[Compensation Eligibility](#)

Select the option that applies to you

- ☒ My flight was delayed or cancelled
- ☐ I missed my connecting flight
- ☐ I lost my baggage or my baggage was damaged
- ☐ I was denied boarding
- ☐ I want clear communication

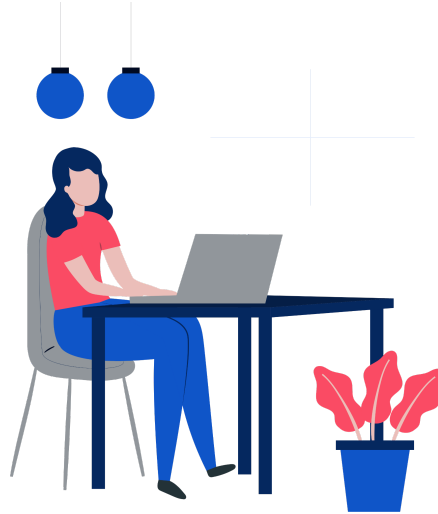
[Make a complaint](#)

- If the user selects the **Compensation Eligibility Checker**, they will be redirected to another section on the website where they will be able to find if they are eligible for compensation or not.

In a scenario where they are eligible, they will also see the complete list of steps they need to follow in order to file for compensation.

If they are not eligible, the tool would tell passengers that they are not eligible for compensation along with the reason why they are not eligible for it, it will also show the corresponding rule from the APPR document to back up the reason.

They will also have the button to read the full version of APPR in order to learn more about the rights that they have.



- This way, the passenger would be able to educate themselves about the rights that they have and this would also help them assess their eligibility for compensation quickly in case they don't have the time to go through the APPR document completely.

Validation

After the solution ideation and creation stage, we conducted a validation interview with participants who had expressed interest in participating in the validation. We also presented our research in the Algonquin Applied Research showcase, where we also received feedback on our solutions as well as areas that should be considered in the live website creation.

Testing Protocols

We created a mid-fidelity compensation checker prototype website to test our solution. Participants were provided with a QR code that directed them to the mid-fidelity prototype website. Upon scanning the code, users could enter details such as their travel destination and travel dates. This process was designed to simulate a real-life scenario where users check their eligibility for compensation due to travel disruptions. Additionally, users could access information about their rights under the Air Passenger Protection Rules (APPR).

To ensure users understood the eligibility tool, we guided them through each step of the process, explaining how eligibility was calculated. This approach allowed us to provide clarity and address any questions during the test. While users independently interacted with the website, we observed their actions to identify any difficulties they encountered. This observation was crucial in evaluating the prototype's usability and determining potential improvements to enhance user friendliness.

Feedback from Validation

The testing sessions provided valuable feedback, highlighting areas for improvement:

- **Airport Lookup:** Users suggested adding a feature that allows searching for airports by city name, as they might not always remember the specific airport name.

- **Navigation:** It was recommended to include the option to go back a step and modify input values in case of errors.
- **Accessibility:** Participants emphasized the need for accessibility features, such as font size adjustments, high contrast mode, and screen reader compatibility.
- **Recent Flight Selection:** Users indicated that having the option to select a recently completed flight, rather than only flights from at least three days ago, would be more practical.
- **Banners:** Users mentioned that having banners in the airport may not be aesthetically pleasing for passengers.

Implemented Feedback

In response to the feedback received, we implemented several improvements to the prototype:

- **Back Button:** A back button was added, allowing users to return to the previous step and correct any errors in the form.
- **Search by City:** We introduced the functionality to search for airports by entering the city name, making it easier for users to locate their departure or arrival airports.

Final Prototype

Our final proposed intervention is titled Project Inform-Me-Right; the aim is to create awareness of the Air Passenger Protection Regulations (APPR), improve communication and give accurate information about flight disruptions to passengers, make the APPR easy to find and understand, and clarify compensation eligibility. The proposed solution covers the major painpoints we discovered from our research.

Awareness:

Awareness of the Air Passenger Protection Regulations (APPR) will be promoted through strategically placed posters in high-traffic areas of the airport where passengers are most likely to notice them. These locations include areas near checkpoints, boarding gates, baggage claim areas, and waiting lounges. By positioning the posters in these prominent spots, they will be more visible to passengers as they move through the airport, ensuring the information is accessible and noticeable to a large number of travelers.

Brochures will be distributed directly to passengers at airline counters, ensuring that all passengers receive the information during check-in or other interactions with airline staff. Additionally, these brochures will be placed in seat pockets on the aircraft, allowing travelers to easily access and read the material during their flight. This provides an opportunity for passengers to learn vital information while seated and likely to have some free time.

Animated graphics will be continuously displayed on screens throughout the airport, including lounges, baggage claim areas, and boarding gates. These engaging visuals are designed to capture passengers' attention and convey important information effectively. The same visuals will also be used as advertisements on the aircraft's entertainment system, ensuring that passengers are made aware of the website both before and during their flight.

Staff training will be implemented to ensure that airline personnel are knowledgeable about the Inform-Me-Right website and its benefits for passengers. As a result, staff at check-in counters and customer service desks will be better equipped to proactively inform passengers about the website.

Social media campaigns will utilize platforms such as Facebook, Instagram, Twitter, and LinkedIn to raise awareness and engage with a broader audience. These campaigns will be designed to capture attention through various engaging formats, including posts, videos, and stories.

Additionally, when customers begin the process of purchasing a ticket, whether online, through a mobile application, or at a physical ticket counter, they will be informed about the existence of the Inform-Me-Right website. This could be done through a message on the booking screen, providing a link to the website, or staff mentioning the availability of the website. The goal is to ensure that passengers are aware of the website and the resources it offers early in their travel planning, enabling them to visit the site to learn more about their rights and other important information they may need while traveling.

Website:

The website will provide a simplified version of the Air Passenger Protection Regulations (APPR), written in user-friendly language. It will also direct passengers to the legal version of the APPR, allowing them to access all relevant information. The website will include an eligibility checker, enabling passengers to determine when they are eligible for compensation and when they are not. This feature aims to reduce confusion and enhance passengers' understanding of the compensation available to them. To further simplify the APPR, we have provided guidelines to help make the information easier for users to find and comprehend.

Simplified APPR Guidelines

A study conducted by the Nielsen Norman Group found that users tend to focus only on topics of personal interest, often overlooking other information unless it is presented with clear and compelling subheadings. This insight highlights the importance of how information is structured and presented, as it greatly influences user engagement. Below are key considerations for simplifying the Air Passenger Protection Regulations (APPR) document:

1. Information Architecture (IA): The website should feature intuitive and clear navigation, allowing users to quickly locate the relevant sections. The content should be organized into distinct topics or sections, making it easier for users to find areas of interest. Once a user selects a specific topic, they should be directed immediately to the relevant page, minimizing the number of clicks and ensuring quick access to necessary information. Poor navigation can frustrate users, leading them to abandon the website. Well-designed IA not only maintains user interest but also increases the likelihood of users staying longer and exploring more content.
2. Prioritize Information: Information should be concise, straightforward, and written in plain language. Key information should be placed at the beginning, and each section should be limited to five lines or fewer to sustain user attention. For instance,
“Standards of Treatment
The regulations establish minimum standards of treatment that airlines have to provide to passengers for delays at departure that are within their control, or within their control and required for safety purposes. After a delay at departure of 2 hours, the airline operating the disrupted flight has to provide: food and drink in reasonable quantities; and a means of communication (e.g., free wifi). If a passenger must wait overnight, airlines have to offer hotel or other comparable accommodation free of charge, as well as free transportation to the accommodation.”

It can be simplified as:

“Assistance

If your flight is delayed for 2 hours due to the airline's fault, they must provide you with adequate food, drinks, and free WiFi. If an overnight stay is required, they must offer free hotel accommodation and transportation to the hotel.”

This approach ensures that the information is clear, direct, and easy to understand, thereby improving user engagement and comprehension.

Compensation Eligibility Checker:

The compensation eligibility checker will be an online tool designed to assist passengers in determining their eligibility for compensation related to flight disruptions. The eligibility checker will simplify the process of understanding compensation rights under the Air Passenger Protection Regulations (APPR).

The website will request essential details about the passenger's journey, including flight information, departure and destination airports, the nature of the disruption encountered, and other relevant factors. Based on this information, the eligibility checker will assess whether the passenger qualifies for compensation. If the passenger is eligible, the tool will provide clear

instructions on the next steps for submitting a compensation claim, ensuring that passengers are well-informed and supported throughout the process. If the passenger is not eligible, the eligibility checker will explain the reasons behind the ineligibility, helping to clarify any misunderstandings about compensation rights. Additionally, the checker will redirect passengers to a simplified version of the APPR section, where they can review the legal documentation if further clarification is needed. This approach ensures that passengers have access to both user-friendly explanations and the formal regulatory text, empowering them with the knowledge they need to make informed decisions.

Our research indicates that many passengers are unaware of their eligibility for compensation, leading them to apply for it without fully understanding the criteria. When these claims are denied, passengers often feel frustrated and unjustly treated, placing blame on the airlines. For example, in situations involving disruptions due to bad weather, airlines are typically not liable for compensation. The eligibility checker we propose aims to address this issue by reducing the number of complaints airlines receive regarding compensation. By guiding passengers through the correct process of applying for compensation, the checker will help them understand the specific reasons behind their ineligibility when applicable. This informed approach will not only streamline the compensation process but also foster a sense of transparency and security among passengers, as they will better comprehend the rules and conditions governing their rights.

The eligibility checker on the website, while valuable, has certain limitations that passengers should be aware of. One notable constraint is the delay in accessing real-time flight data. Since the flight information necessary to determine eligibility might not be immediately available, passengers could experience a waiting period before they can check if they are entitled to compensation. This delay could cause frustration, especially in situations where passengers are eager to know their rights during disruptions. Another limitation involves network accessibility. When traveling, passengers may not always have reliable access to the internet, especially in areas with poor connectivity or in-flight. This lack of access could hinder their ability to use the eligibility checker at crucial moments, such as during or immediately after a disruption. As a result, passengers might miss the opportunity to promptly understand their eligibility status or take timely action to claim compensation. Additionally, the checker's reliance on accurate input from passengers, such as correct flight details and the nature of the disruption, is crucial for providing the right information. Any errors or omissions in this data could lead to incorrect results, further complicating the process and potentially leading to misunderstandings about eligibility. Finally, while the eligibility checker provides a clear pathway for understanding compensation rights, it does not address all possible scenarios or the nuances of individual cases. Some complex situations may require further clarification or direct communication with the airline, which the checker cannot fully accommodate. This limitation underscores the importance of passengers being well-informed and prepared to seek additional information if needed.

Reasons for Website Solution

1. It is easy to access, and it will be available at any time and place; passengers will be able to use their devices to read and understand their rights at anytime or place, which will also give them confidence. SMEs said that it would be great if passengers could quickly know about their rights.
2. There is an existing simplified website in the European Union (EU) for EU passenger rights, but it doesn't involve the Canadian APPR. We understood the need for a website for the Canadian APPR.
3. It would also ensure information given to passengers is always accurate. The website can always be updated or modified when changes are made to a policy or when the policy is updated. That will ensure passengers always receive updated information as quickly as possible.
4. It will also empower passengers to know what steps to take and what rights apply for their situation.
5. Having a website will reduce the amount of calls and complaints at the customer service desk. Passengers will not need to repeatedly ask the staff and customer service about the flight disruption because they will be getting the initial updates through the website.
6. It also makes it easier to store and share a large amount of information with passengers.

Short-Term, Medium, and Long-Term Goals

After completing quantitative and qualitative analysis, we discovered our key insights, experience principles, and how we might make statements that lead us to begin our solution ideation. We came up with different possible solutions to create awareness and make the APPR easier to understand; ultimately, we separated the solutions into short-term, medium-term, and long-term options;

Short-Term Goals:

1. Develop and Launch the Website: Create a user-friendly website that simplifies APPR and includes a compensation eligibility checker. Ensure the site is accessible and easy to navigate.
2. Create Awareness: Design and distribute posters, brochures, and airport graphics to raise awareness about APPR and direct passengers to the website.
3. Collect Initial Feedback: Gather feedback from users to refine the website and awareness materials.

Medium-Term Goals:

1. Expand Awareness Campaign: Increase the distribution of awareness materials to more airports and travel agencies. Consider digital campaigns on social media platforms.
2. Enhance Website Features: Incorporate additional tools, such as FAQs, live chat support, and multi-language support, based on user feedback.
3. Regulating airlines to integrate the website into their communication channels, such as booking confirmations and in-flight announcements.

Long-Term Goals:

1. Monitor and Improve: Continuously monitor user feedback, update the website, and refresh awareness materials to maintain relevance and effectiveness.
2. Integrate with Travel Platforms: Work towards integrating the compensation checker into major travel booking platforms and airline apps.

What are the benefits for the client and the users?**For the Client:**

This solution offers the client—whether an airline, airport, or regulatory body—a streamlined approach to bridge the communication gap between the services provided and passenger expectations. By implementing a website with a simplified APPR and a compensation eligibility checker, the client can enhance transparency, reduce customer complaints, and demonstrate a commitment to passenger rights. The awareness materials, including posters, brochures, and airport graphics reinforce the client's dedication to clear communication and customer satisfaction, ultimately leading to improved passenger experiences and potentially reducing the burden of handling compensation disputes.

For the Users:

For passengers, this solution provides a clear and accessible way to understand their rights and compensation options during disrupted connecting flights. The website simplifies complex regulations, allowing users to quickly assess their eligibility for compensation. The awareness materials distributed in airports help ensure passengers are informed even before they experience a disruption, reducing confusion and frustration. Overall, the solution enhances the travel experience by giving passengers the knowledge and tools they need to navigate disruptions more confidently and effectively.

Conclusion

In conclusion, the research highlights the gap in communication between passengers and their airlines, especially when it comes to the rights of the passengers in relation to interruptions of flights caused by missed connections. This gap does not only make such disruptions harder for travelers but also affects their entire travel experience and their trust in the air travel industry. Flight delays and missed connections are complex issues that involve various factors like obstruction of operations, inadequate communication, and emotional impacts for the passengers.

The implications of these findings are profound. Airlines that prioritize clear communication, operational efficiency and customer satisfaction are likely to see an increased passenger loyalty, good brand reputation and competitive advantage in the market. For passengers, these improvements mean a more seamless and less stressful travel experience, where they feel valued and informed, even in the face of disruption.

Some of the key solutions that our study proposes are proactive communication strategies. Real-time updating of information, clear information regarding passenger rights, and proper practices of customer engagement at times of flight disruption are very important for an airline to reduce most of the confusions and frustrations felt by passengers. This approach not only helps passengers make better decisions but also instills trust and feelings of reliability in airlines. At the heart of these solutions is the development of a dedicated website, which serves as one of the most important tools in bridging the gap between airlines and the passengers. By offering a centralized platform where passengers can easily access the information they need, thereby avoiding confusion and reducing stress among passengers to enable them to make informed decisions in the event of travel disruptions.

Looking ahead, these solutions can help to realize a more resilient and passenger-centric system of air travel. Equipped with real-time data, predictive analytics and state-of-the-art communication devices, airlines will be better positioned to foresee and proactively limit the effects of potential disruptions. This not only supports operational efficiency but also ensures a more enjoyable and hassle-free journey for passengers. Further, exploring how different passenger demographics interact with the website will be essential for tailoring its features to meet diverse needs.

Moreover, it is important to manage the emotional needs of passengers during disruptions in order to maintain their loyalty. Updating, empathizing, and guiding the passengers on matters involving trouble or hassle can turn quite a few negative experiences into chances of increasing further loyalty.

In the future, these solutions will become highly critical as the industry goes through an increasingly complex and dynamic environment. Given the growing demand for air travel, changing passenger expectations, and increasing competition between airlines, any airline that focuses much attention on effective communication, operational efficiency, and customer satisfaction will be well-placed to compete in this market.

Ultimately, bridging communication gaps, smoothing the passenger journey, and optimizing operations will be important not only in easing the current travelling experience but also in setting the stage for a more sustainable and resilient future of air travel.

Supporting Documents for Case Study Report

Appendix A: Full reference list for literature review

Ali, H., Guleria, Y., Alam, S., & Schultz, M. (2019, December). A Passenger-Centric model for reducing missed connections at low-cost airports with gates reassignment. *IEEE Journals & Magazine | IEEE Xplore*, 7, 179429-179444. <https://doi.org/10.1109/ACCESS.2019.2953769>

The study presents a model based on a passenger-centred analysis of various operational factors affecting missed connections for self-connecting passengers, focusing on low-cost carriers at Singapore Changi Airport Terminal 4. (Ali et al, 2019, pp. 179429-179430). The study aims to assess the impact of stochastic delays, minimum connection times, and flight turnaround times on the number of missed connections among low-cost carriers operating in budget terminals (Ali et al, 2019, p. 179432). The proposed passenger-centric model comprises four major components: operations, passenger flows, disruption patterns, and infrastructure (Ali et al, 2019, p. 179432). The three important operating factors viz. variations of Turn Around Time (TAT), Minimum Connection Time (MCT), and arrival delays are analyzed to understand their relationships (Ali et al, 2019, p. 179432). For optimal gate assignments minimizing passenger transit time, a heuristic search algorithm, which is based on the Tabu-search algorithm is used (Ali et al, 2019, p. 179434).

It was observed that the passengers missed connections which have less than 45 minutes of buffer time. (Ali et al, 2019, p. 179440). For the convenience of passengers, a 60-minute buffer time between connecting flights is recommended. (Ali et al, 2019, p. 179440). The experiments indicate that increasing TATs reduces the probability of travellers missing connections. Higher TATs allow aircraft to spend more time on the ground, making connections more convenient for passengers. (Ali et al, 2019, p. 179441). Reducing delays can significantly reduce missed connections, even if they are only 30 minutes apart (Ali et al, 2019, pp. 179441-179442). Improving TAT and MCT and decreasing delays can greatly decrease missed connections (Ali et al, 2019, p. 179442). Transfer passenger missed connections can be prevented for almost all the flights by maintaining flight TAT at 50 minutes, MCT at 60 minutes, and arrival delays within 70% of the current delay spread (Ali et al, 2019, p. 179443).

Baker, D. (2013). Service Quality and Customer Satisfaction in the Airline Industry: A Comparison between Legacy Airlines and Low-Cost Airlines. *American Journal of Tourism Research*, 2(1), 75-76. <https://doi.org/10.11634/216837861403317>

The study explores service quality and customer satisfaction in the airline industry. The article centred around the differences between legacy and low-cost carriers (LCCs), identifying the critical factors for the operational structures and competitive strategies for both categories. He suggested that understanding how the legacy airlines operated compared to the low-cost carriers would enable the airlines to compete. He focused specifically on differences such as cost structure, market share, infrastructure, and customer service (Baker, 2013, pp. 75-76). Although both airlines deliver flights, he pointed out the differences between the two. LCCs are characterized by their market value, point-to-point networks, their focus on being low-cost, and thin service delivery. These findings are seen as showing that, although LCCs are gaining market share in mature markets such as Europe and North America, their penetration of more rapidly growing markets such as Asia and Latin America remains elusive (Baker, 2013, pp. 75-76). The same paper is scathing in its critique of the potential for further increases in LCCs' market penetration and emphasizes the role of low-cost carriers in upending the established thinking around the role of airports in terms of public and private investment and regulatory planning opportunities. The upshot is that in this continuing and 'disruptive' transformation in the airline business, generations of thinking about what works and what doesn't need to unlearn its orthodoxies and embrace the new paradigm – especially amidst LCC growth.

Furthermore, because the air transportation industry is heavily customer-oriented, Baker goes on to demonstrate how airlines' success is related to achieving high customer satisfaction: Importantly, the key to success in any industry is meeting or exceeding the expectations of one's customers. When consumers' expectations of service are not met, customers may either turn to another brand for a service (for example, they may consider flying a different airline) or they may accept substandard output (they may believe that a specific level of service, such as the attitude of a flight attendant, is "ever present in the air travel industry") (Baker, 2013, pp. 75-76). As such, Baker argues that when service offerings are matched to the realistic expectations of consumers and service complaints are dealt with properly, airlines can improve the differentiation of their brand and shape passengers' positive behavioural intentions. Overall, the study offers useful implications for understanding airlines – both the legacy carriers and the low-cost carriers – for marketing research, and for the strategies and practices to improve service quality and customer satisfaction in the airline business.

Bogicevic, V., Bujisic, M., Bilgihan, A., Yang, W., Cobanoglu, C. (2017). The impact of traveler-focused airport technology on traveler satisfaction. *Technological Forecasting and Social Change* 123, 351-361. <https://doi.org/10.1016/j.techfore.2017.03.038>

The level of trust that airline passengers have in the airline industry is quite low because airports do not meet travelers' expectations due to the incompetence of airport staff, long waiting times, poor navigation systems, a lack of information, and inconvenient flight

schedules (Bogicevic et al, 2017, p. 1). Passengers expect to receive valuable information and efficient processing from airport staff, precise online information, and even self-service options; therefore, technology has become an integral part of air travel (Bogicevic et al, 2017, p. 1). There is evidence showing that technology has helped to reduce the waiting time for different points like check-in, baggage drop, and so on. The study focuses on finding the relationship between airport technologies and traveler satisfaction, and how technologies may influence the positivity of traveler satisfaction (Bogicevic et al, 2017, p.2). The study makes use of a survey research method to gather data directly from travelers about their experiences and perceptions of airport technologies, especially because of the large number of respondents, the survey was the most suitable method (Bogicevic et al, 2017, p.3-4).

Four regression paths were tested for significance, each representing one of four hypotheses and the model result supported the four hypotheses, showing that there is significance (Bogicevic et al, 2017, p. 6-7). The study concluded that the use of airport technologies, namely self-service kiosks, mobile apps, and charging stations, has a positive effect on travelers' confidence, enjoyment, and overall satisfaction with their airport experience (Bogicevic et al, 2017, p. 7-9). By investing in such technologies that prioritize the needs of travelers, airports can enhance passenger satisfaction and create a more pleasant and efficient travel environment. The study mentions the limitations, it is important to note that respondents answered the survey questions based on their memory of previous travel experience therefore there is a possibility of inaccurate information and then it goes on to give future research suggestions.

Borsky, S., & Unterberger, C. (2019). Bad weather and flight delays: The impact of sudden and slow onset weather events. *Economics of Transportation*, 18, 10–26.

<https://doi.org/10.1016/j.ecotra.2019.02.002>

This article investigates how bad weather affects airline departure delays. It focuses on sudden onset events such as precipitation and wind, as well as slow onset events like temperature fluctuations (Borsky & Unterberger, 2019, pp. 10-11). The paper investigates the cost of delays, which were found to be significant. The study explains how weather affects air travel and suggests the aviation industry on efficiency improvement and cost reduction of delays. (Borsky & Unterberger, 2019, p. 11). The researchers implemented statistical methods to calculate weather related delays by collecting data from 2.14 million flights at ten major airports (Borsky & Unterberger, 2019, p. 11).

The study analyzes the impact of sudden onset events on departure delays and the findings indicate that bad weather conditions significantly increase departure delays. (Borsky & Unterberger, 2019, pp. 15-16). Rain and snow contribute to delays ranging from 10 to 23 minutes. Wind effects are minor, which last for 1 to 3 minutes. Frost events increase delays by 2 to 3 minutes. The heat effects are negligible. (Borsky &

Unterberger, 2019, pp. 18-20). Improving airline performance during weather circumstances could aid in mitigating the effects of weather induced delays (Borsky & Unterberger, 2019, pp. 20).

Bratu, S., & Barnhart, C. (2005). An analysis of passenger delays using flight operations and passenger booking data. *Air Traffic Control Quarterly*, 13, 4-18.
<https://doi.org/10.2514/atcq.13.1.1>

Bratu and Barnhart (2005) look closely at how scheduling airline delays for passengers affects different types of travellers for a large US hub-and-spoke airline using data on flight operations and passenger bookings for an entire month, August 2000. For all their detailed work, the dataset for their analysis was reasonably typical at the time, based on 307,675 planned itineraries for 2.56 million passengers (pp. 4-5). The authors worked with this legacy data, using Passenger Disruption Control (PDC) techniques to help find new itineraries for the affected passengers, and to re-schedule passengers in order of the time they were disrupted, ensuring a first-disrupted-first-recovered service policy (pp. 4-5).

For each flight, the analysis also compares passenger delays to flight leg delays and finds wide disparities, with passenger delays being 1.7 times higher than flight leg delays, with an average of 25 minutes and 15.4 minutes, respectively (pp. 14-15). Disrupted passengers experience extremely long delays, especially affected passengers such as those with flight cancellations and missed connections, and failure of flight-based metrics to capture, from a passenger perspective, the full meaning of schedule performance. The study then considers why passengers are disrupted: they group their passenger delay data into categories such as local versus connecting passengers, whether disruptions happen during specific times of the day, and the metrics for flight scheduling design affecting passenger delay (pp. 16-18). By subjecting a boatload of data to careful analysis and empirical exploration, Bratu and Barnhart demystified a vital part of the airline system – and prepared for the day when we can resolve those pesky scheduling problems.

Britto Rodrigo, Martin Dresner, Augusto Voltes (2012). The impact of flight delays on passenger demand and societal welfare, *Transportation Research Part E: Logistics and Transportation Review*, 48(2), 460-469. <https://doi.org/10.1016/j.tre.2011.10.009>

The paper by Britto et al. (2012) talks about the impact of flight delays on passenger demand and airfares. The authors address their concern over growing flight delays and their impacts on both consumers and producers. By analyzing data on flight delays and scheduled block times, the study helps us understand how delays can influence passenger decisions and airline costs. The authors consider factors such as consumer behaviour when their flight is delayed, the effects on airline operations, and the resulting

changes in airfares. With the help of some estimations and welfare calculations done in the past, the researchers assess the overall impact of flight delays on society's welfare. (Britto et al., 2012, p. 2).

Britto et al. (2012) found that flight delays have a significant negative effect on passenger demand leading to increased airfares. The analysis reveals that delays not only reduce consumer welfare but also result in higher costs for airlines. They also estimated that a reduction in delays could potentially lead to substantial gains for both passengers and airlines, emphasizing the economic importance of addressing this issue. By highlighting the relationship between flight delays, passenger demand, airfares, and welfare impacts, the study lights on the need for effective strategies to mitigate delays and improve efficiency in the aviation sector. Overall, the findings suggest that there is a need for improvements aimed at reducing flight delays that can benefit both consumers and producers in the airline industry. Also, the authors suggest that the cost of paying for infrastructure improvements to reduce delays should be shared but airlines should be assessed for the majority of the costs since they are the main beneficiaries of delay reductions. (Britto et al., 2012, p. 7).

Castaing, J., Mukherjee, I., Cohn, A. M., Hurwitz, L., Nguyen, A., & Müller, J. J. (2016). Reducing airport gate blockage in passenger aviation: Models and analysis. *Computers & Operations Research*, 65, 189–194. doi: <http://dx.doi.org/10.1016/j.cor.2014.02.011>

In their article, Castaing et al. (2016b) address the issue of gate blockage in commercial aviation, which arises when an arriving flight is delayed due to the previous aircraft still occupying the gate. Gate blockage leads to passenger delays, missed connections, and increased fuel burn, among other negative impacts (p. 189). The authors analyze historical data to understand the frequency and extent of gate blockage and propose models to optimize gate assignments under stochastic conditions. By incorporating information about variability in flight arrival and departure times, they develop mixed integer programming (MIP)--based models to minimize gate blockage occurrences. Through empirical analyses using real-world data, they demonstrate the computational tractability and potential benefits of their proposed approach (pp. 189-190).

The study's methodology involves a historical analysis of gate blockage frequency and patterns, focusing on the domestic operations of a major U.S. carrier. The authors estimate gate blockages by reverse-engineering available flight data and calculate metrics such as the frequency and duration of gate blockages across different periods and airport stations. Results show that gate blockages affect roughly 5% of flights, with varying durations, but even short blockages can have significant impacts, particularly on connecting passengers (p. 191). The analysis highlights the importance of considering gate blockages in planning processes to minimize disruptions and optimize airline operations. This research contributes to the literature on airline gate planning by assessing the impact of stochasticity on gate blockage and proposing MIP-based

approaches to mitigate this impact, thus offering insights for improving airport efficiency and passenger experience (pp. 191-192).

Cattaneo, M., Malighetti, P., Paleari, S., & Redondi, R. (2019). Manage with care: the frailty of self-connections in the European airport network. *European Transport Research Review*, 11(1), 1-3. <https://doi.org/10.1186/s12544-019-0387-z>

Cattaneo et al. (2019) delve into the intricacies of self-connectivity within the European air transportation network, a burgeoning trend propelled by the expansion of low-cost carriers (LCCs) and the advent of systems like ViaMilano and GatwickConnect (Cattaneo et al., 2019, p. 2). Despite offering passengers alternative routing options and flexibility, the authors underscore the challenges airlines and airports face in sustaining and effectively managing self-connections (Cattaneo et al., 2019, p. 2). Issues such as the dwindling number of indirect connections over time and the logistical hurdles in implementing dedicated services for self-connecting passengers cast doubts on the long-term viability of this approach (Cattaneo et al., 2019, p. 2). Through an empirical analysis, the study meticulously examines the performance of self-connections vis-à-vis alliance-based and online connections, shedding light on factors like travel times, frequency of connections, and the feasibility of return trips (Cattaneo et al., 2019, p. 2). The findings of the research reveal intriguing insights into the dynamics of self-connectivity, painting a nuanced picture of its strengths and limitations (Cattaneo et al., 2019, p. 2). While self-connections tend to entail longer travel times due to less efficient routing and lack of coordination between airlines, they offer a substantial number of viable connections, particularly within Europe (Cattaneo et al., 2019, p. 2). However, their lower frequencies compared to alliance-based options pose challenges for passengers seeking more reliable and convenient travel arrangements (Cattaneo et al., 2019, p. 2). Moreover, the study evaluates the feasibility of return trips under different scenarios, providing valuable insights for passengers and industry stakeholders alike (Cattaneo et al., 2019, p. 2). Ultimately, the research contributes to a better understanding of the evolving landscape of air transportation in Europe, informing strategic decisions regarding the promotion and management of self-connections in the years to come (Cattaneo et al., 2019, p. 2).

Cheng, S., Ya-Ping, Z., Hao, S., Liu, R., Luo, X., & Luo, Q. (2019). Study of flight departure delay and causal factor using spatial analysis. *Journal of Advanced Transportation*, 2019, 1–11. <https://doi.org/10.1155/2019/3525912>

The study investigates delay aggregation patterns by measuring and quantifying the spatial dependence of delays (Cheng et al, 2019, p. 1). This research uses data from Bureau of Transportation Statistics and IATA to validate the suggested asymmetric Bayesian logit model and uncover novel factors influencing arrival delay probability. The

study examines departure delay as a geographically distributed quantity structured by hour and day, rather than a statistical problem (Cheng et al, 2019, p. 2). Spatial analysis is a quantitative method for studying spatial variables. This method investigates flight departure delay distribution patterns and causal elements. Delay is believed to be a spatially distributed parameter (Cheng et al, 2019, pp. 2-3) The causing factors are categorized into three groups: high frequency and low effect, low frequency and high effect, and low frequency and low effect (Cheng et al, 2019, p. 4).

Flow control, airline factors, route restrictions, and weather conditions at the previous airport caused most of the departure delays (Cheng et al, 2019, p. 4). External variables, such as sudden bad weather conditions, are likely to produce flight departure delays on a large scale (Cheng et al, 2019, p. 5). Weather conditions influence flight departure delays at the preceding and departure airports, flow control, and the number of scheduled departures. Technical failures and route restrictions have a crucial role as well. Adverse weather has a greater influence than flow control, therefore attempts to reduce delays focus primarily on proactive weather forecasting and dynamic changes to changing conditions (Cheng et al, 2019, pp. 8-10).

Fernandes, N., Moro, S., Costa, C. J., & Aparício, M. (2020). Factors influencing charter flight departure delay. *Research in Transportation Business & Management*, 34, 100413.
<https://doi.org/10.1016/j.rtbm.2019.100413>

The study uses data mining techniques to examine the key causes of charter flight departure delays (Fernandes et al, 2019, p. 1). This study highlights factors impacting flight delays in the crucial European market (Fernandes et al, 2019, p. 2). This investigation was conducted on charter flights using data-based sensitivity analysis to identify the most relevant features associated with aircraft delays (Fernandes et al, 2019, p. 2). The empirical research is based on data from a European airline operating flights globally. The goal is to anticipate if a flight is delayed by more than 15 minutes and analyze the reasons for the delays (Fernandes et al, 2019, p. 3). The selection of features, including season, aircraft size, ad-hoc flight, and longitude is based on literature (Fernandes et al, 2019, p. 3). the neural network (NN), support vector machine (SVM), and random forest (RF) these three modelling techniques were chosen (Fernandes et al, 2019, p. 4).

The aggregate relevance of all features connected to the aircraft's previous flight is around 48% (Fernandes et al, 2019, p. 6-7). If a flight is delayed for two registered reasons, there is a more than 30% chance of the flight being delayed again on the current trip (Fernandes et al, 2019, p. 7). The probability of a flight delay varies depending on whether it is an ad-hoc or scheduled flight. Although the difference is only approximately 0.05, ad-hoc flights are less likely to be late than planned flights (Fernandes et al, 2019, p. 8). The reason for the previous flight delay was just 2%

relevant. However, the feature indicating if the preceding flight had many explanations accounted for around 33% relevance (Fernandes et al, 2019, p. 9).

Henriques, R., & Feiteira, I. (2018). Predictive modelling: Flight delays and associated factors, Hartsfield–Jackson Atlanta International Airport. *Procedia Computer Science*, 138, 638–645. <https://doi.org/10.1016/j.procs.2018.10.085>

The project seeks to predict arrival delays at the Hartsfield–Jackson Atlanta International Airport by applying various data mining approaches (Henriques & Feiteira, 2018, p. 639). To predict arrival delays at ATL, several approaches were used, including the use of different sampling techniques, the inclusion and exclusion of outliers, the application of different attribute selection methods, the inclusion and exclusion of two variables, and, finally, the application of three different supervised machine learning algorithms (Henriques & Feiteira, 2018, p. 639). To estimate flight delays, relevant data was carefully selected, cleaned to remove missing values and outliers, and then optimized through data transformation. The three algorithms utilized to achieve this were decision trees, random forests, and multilayer perceptrons. ROC curves and F-measures are useful evaluation tools that stress the importance of accurate data preparation and algorithm selection in predictive analysis. (Henriques & Feiteira, 2018, pp. 640-642). The comparison section compares Decision Trees, Random Forests, and Multilayer Perceptron for predicting flight delays. Multilayer Perceptron exceeds all three with 85% accuracy using SMOTE, outlier removal, and the GainRatio technique (Henriques & Feiteira, 2018, p. 643). Future recommendations include investigating other approaches and tools for improvement, suggesting regression extensions, and creating a more comprehensive prediction model to overcome study constraints (Henriques & Feiteira, 2018, pp. 643-644).

Jiang, Y., Li, S., Huang, J., & Scott, N. (2019). Worry and anger from flight delay: Antecedents and consequences. *International Journal of Tourism Research*, 22(3), 289–294. <https://doi.org/10.1002/jtr.2334>

Jiang et al. (2019) explore the complex interplay between emotions and flight delays, shedding light on the antecedents and consequences of worry and anger experienced by passengers. Drawing attention to the escalating concern over airline punctuality, particularly in China where flight delays have become increasingly prevalent, the study underscores the need for a deeper understanding of how these negative emotions arise and their potential ramifications (Jiang et al., 2019, p. 289). By examining the cognitive appraisal theory (CAT) as a theoretical framework, the researchers delve into the subconscious evaluations passengers make when faced with flight delays, elucidating the dimensions of appraisal that contribute to the elicitation of specific emotions such as worry and anger (Jiang et al., 2019, p. 290). Through empirical analysis, the study not

only confirms the validity of CAT in understanding the emotional responses of passengers but also extends the literature by investigating the distinct effects of worry and anger on passengers' coping mechanisms and behavioural intentions (Jiang et al., 2019, p. 290).

The findings of the study underscore the critical role of cognitive appraisal dimensions such as goal congruence, agency, and certainty in shaping passengers' emotional responses to flight delays (Jiang et al., 2019, p. 290). While goal incongruence tends to evoke both worry and anger, the appraisal of agency and certainty further differentiates these emotions, highlighting the nuanced mechanisms underlying passengers' emotional experiences during flight delays (Jiang et al., 2019, p. 291). Moreover, the study reveals the significant impact of worry and anger on passengers' coping strategies and behavioural intentions, with anger being linked to more assertive responses such as switching airlines and lodging complaints, while worry tends to result in more passive coping behaviours (Jiang et al., 2019, p. 291). These findings contribute to a deeper understanding of how passengers navigate and respond to the challenges posed by flight delays, offering valuable insights for airlines and other stakeholders in managing customer emotions and enhancing the passenger experience (Jiang et al., 2019, p. 291).

Hai Jiang and Cynthia Barnhart (2009), *Dynamic Airline Scheduling*, *Transportation Science* 2009 43(3), 336-354. <https://doi.org/10.1287/trsc.1090.0269>

Jiang and Barnhart's (2009) paper "Dynamic Airline Scheduling" discusses the difficulty airlines have in maximizing profitability while meeting fluctuating passenger demand through flight schedule optimization. The study emphasizes the necessity of dynamic scheduling techniques to modify schedules in real time and focuses on the effects of demand on airline scheduling and profitability. To better match capacity with demand, the authors suggest a dynamic scheduling model that reoptimizes aircraft schedules during the passenger booking process (Hai Jiang and Cynthia Barnhart, p.1).

Jiang and Barnhart (2009) also provide evidence in their study regarding the efficiency of dynamic scheduling in enhancing airline operations. A weekly timetable that is customized for each day of the week can result in better revenues than a static daily schedule, according to a case study that examines the variability in daily passenger requests for a major U.S. airline over 4 weeks (Hai Jiang and Cynthia Barnhart, p. 44). The findings imply that airlines can increase overall profitability and better capture fluctuations in demand with the use of dynamic scheduling. The study highlights the potential advantages of applying dynamic scheduling tactics in the airline industry and emphasizes how crucial it is to modify aircraft plans in response to shifting demand trends.

Kim, N., & Park, J. (2016). A study on the impact of airline service delays on emotional reactions and customer behavior. *Journal of Air Transport Management*, 57, 19–25.
<https://doi.org/10.1016/j.jairtraman.2016.07.005>

This study examined how service delays affect airline passengers' behavioural intentions based on their positive and negative emotional reactions (Kim & Park, 2016, pp. 19-20). According to the study, a service delay refers to the amount of time passengers have to wait for air transport service (Kim & Park, 2016, p. 20). Experiencing a service delay may cause an immediate emotional response (Kim & Park, 2016, p. 20). Social evaluation theories categorize emotional responses into three domains: acceptance, denial, and non-commitment. There are categories of opinions within an acceptance domain. The denial domain refers to customers who are unwilling to wait, whereas the non-commitment domain refers to customers who are unsure whether to accept or reject a wait (Kim & Park, 2016, pp. 20-21). The study used empirical data to form hypotheses (Kim & Park, 2016, p. 21). A survey questionnaire was constructed to contain the six constructs included in the hypotheses, which were service delay, anger, uncertainty, acceptability, repurchase intention, and negative word-of-mouth (Kim & Park, 2016, p. 21).

The hypothesis verification showed statistical significance except for two hypotheses (Kim & Park, 2016, p. 22). Service delays significantly affected anger, uncertainty, and acceptability. It contributed to negative emotions, whereas anger and uncertainty were positively influenced.

Anger negatively affected repurchase intention but positively affected negative word-of-mouth. Uncertainty did not have a major impact on repurchase intentions or negative word-of-mouth. Acceptability influenced both repurchase intention and negative word-of-mouth. (Kim & Park, 2016, pp. 22-23). Airlines should take measures to manage passengers' emotional responses to prevent negative word-of-mouth and discourage negative repurchasing intentions owing to service issues (Kim & Park, 2016, pp. 24-25).

M. Kešel'ová and P. Hanák (2019), Risk and opportunities in the process of flight delay, 2019 New Trends in Aviation Development (NTAD), Chlumec nad Cidlinou, 87-91.
<https://doi.org/10.1109/NTAD.2019.8875614>

The study "Risk and Opportunities in the Process of Flight Delay" by Kešečová and Hanák (2009) examines the challenges faced by the aviation sector due to flight delays and emphasizes the importance of risk and process management. The authors discuss how flight delays can result in monetary losses, angry passengers, and operational interruptions. In addition to emphasizing the need for prompt action to ensure the continuation of air traffic, they provide a risk management-based technique to identify and address the underlying causes of flight delays (Kešečová and Hanák, p. 2). The authors present a rigorous methodology that includes problem identification, analysis,

solution implementation, and resolution to handle flight delays successfully. Furthermore, Kešečová and Hanák's research highlights the need for process management and risk assessment to minimize the risks associated with flight delays (Kešečová and Hanák, p. 3). Airlines can use tools like risk registers and SIPOC diagrams to identify crucial performance indicators and assess the impact of delays on various aspects of their operations. The proposed methodology offers airlines a structured framework to save costs, enhance stakeholder satisfaction, and address aircraft delays in advance. The report provides useful details regarding the difficulties in managing flight delays as well as a methodical plan for lowering risks and grabbing opportunities in the aviation industry.

Shan Lan, John-Paul Clarke, Cynthia Barnhart (2006), Planning for Robust Airline Operations: Optimizing Aircraft Routings and Flight Departure Times to Minimize Passenger Disruptions, 40(1). <https://doi.org/10.1287/trsc.1050.0134>

Lan et al. (2006) go into great detail on the subject of optimizing aircraft routings and flight departure times to minimize passenger disruptions in the airline industry. The authors' paradigm for robust airline operations planning aims to proactively handle schedule disruptions by strengthening the scheduling process. They stress how important it is to build resilience into airline schedules to mitigate the impact of disruptions during the planning stage. By focusing on improving aircraft routings and departure times, the study seeks to increase operational efficiency and customer satisfaction in the event of disruptions (Lan et al., p. 16).

In this paper, Lan et al. (2006) offer a string-based formulation for robust aeroplane maintenance routing that helps to lower the total expected propagating delay. The authors demonstrate how aircraft rerouting and purposeful scheduling of flight legs within brief periods can reduce delays and inconveniences for passengers. To strengthen operational resilience and lessen the effects of delays, the study emphasizes the necessity of anticipatory planning in the industry. By presenting proof-of-concept results and discussing the implications of their modelling technique, Lan et al. (2006) offer significant insights into the resilience and efficiency optimization of airline operations (Lan et al., pp. 21-23).

Lettovský, L. (1997). Airline operations recovery: An optimization approach. Retrieved from Georgia Institute of Technology Institutional Repository, 1-30.
<https://smartech.gatech.edu/handle/1853/24326?show=full>

Lettovský (1997) examines the optimization of airline operations recovery across multiple pages, providing a detailed analysis of strategies to address disruptions in the aviation industry. From page 1 onwards, the study delves into the complexities of mitigating disruptions such as flight delays and cancellations, emphasizing the application of

optimization techniques to minimize their impact on airline operations. Lettovský's research offers valuable insights into the development of efficient strategies for coping with unforeseen events and restoring normal operations within airlines (Lettovský, 1997, pp. 1-15).

Continuing through subsequent pages, Lettovský's work explores the theoretical frameworks and practical methodologies for enhancing the resilience and reliability of airline operations. By integrating optimization techniques into the recovery process, the study provides a valuable foundation for addressing the operational challenges faced by airlines and advancing the efficiency of their response to disruptions (Lettovský, 1997, pp. 16-30). This comprehensive examination extends across various sections of the research, offering a holistic perspective on the optimization of airline operations recovery.

Mitsokapas, E., Schäfer, B., Harris, R. J., & Beck, C. (2021). Statistical characterization of airplane delays. *Scientific Reports*, 11(1). <https://doi.org/10.1038/s41598-021-87279-8>

The study focuses on airline delays at UK airports between 2018 and 2020. Flight delays significantly reduce customer satisfaction and may drive customers to choose a different airport or airline in the long run (Mitsokapas et al, 2021, p. 1). To illustrate the effect the lockdown had on the delay distribution, The data is gathered from the top 5 UK airports with a specific focus on London Heathrow Airport during the 2020 COVID-19 pandemic (Mitsokapas et al, 2021, p. 2).

The delay statistics for COVID-19 were shifted to the left, suggesting overall smaller delays (including more negative delays), while the basic shape of the distribution did not change significantly (Mitsokapas et al, 2021, pp. 2-3).

Several factors influence the delay distribution for each airport or airline. Airline policies, flight routes, technological problems, and documentation errors cause 27% of delays. Overseas flights are more vulnerable to wind, unpredictable weather, and military exercises (Mitsokapas et al, 2021, p. 4). COVID-19 led to a significant drop in global flight traffic, resulting in many flights arriving early (Mitsokapas et al, 2021, p. 5). The influence of the COVID-19 pandemic on airline delay statistics is significant, particularly at Heathrow Airport (LHR). Despite travel restrictions and quarantine precautions, LHR experienced surprisingly low average delays during the pandemic. However, there were still cases of significant delays. These results imply that, in certain COVID-19 settings with fewer flights, airports can perform better in specific areas, such as mean delay, than under normal conditions (Mitsokapas et al, 2021, pp. 8-9).

Pternea, M., & Haghani, A. (2019). An aircraft-to-gate reassignment framework for dealing with schedule disruptions. *Journal of Air Transport Management*, 78, 116–122. <https://doi.org/10.1016/j.jairtraman.2019.01.005>

Pternea and Haghani (2019) present a comprehensive framework for aircraft-to-gate reassignment to manage schedule disruptions effectively. Their study, outlined in the *Journal of Air Transport Management* (p. 116), examines the intricacies of gate assignments, emphasizing the need for a flexible approach to accommodate changes in flight schedules. By analyzing the challenges airlines face during unexpected events (p. 116), the authors propose a systematic framework designed to optimize gate reassignments, providing airlines with a strategic tool to enhance operational resilience. The research offers valuable insights into the complex dynamics of airport operations and underscores the critical role of gate assignments in mitigating the effects of schedule disruptions (p. 118). By offering a systematic approach to aircraft-to-gate reassignment, Pternea and Haghani contribute to the advancement of airline management practices, addressing practical challenges while emphasizing the importance of proactive strategies in maintaining operational efficiency and customer satisfaction amidst disruptions.

Gregory D. Roach, Renée M. A. Petrilli, Drew Dawson & Nicole Lamond (2012) Impact of Layover Length on Sleep, Subjective Fatigue Levels, and Sustained Attention of Long-Haul Airline Pilots, *Chronobiology International*, 29(5), 580-586, <https://doi.org/10.3109/07420528.2012.675222>

The Roach et al. study looks at how long layovers affect sleep, reported fatigue, and sustained focus in long-haul airline pilots (2012). The researchers measured the level of tiredness and the quality of sleep using wrist activity monitors, self-report sleep diaries, and the Samn-Perelli exhaustion Checklist. They also measured sustained attention using the psychomotor vigilance test and investigated the effects of short and lengthy layovers on pilot performance and well-being using mixed-model regression analysis. The goal of the study was to address the critical issue of pilot fatigue, which the aviation industry is very concerned about due to potential consequences for operational effectiveness and safety.

The study's findings demonstrated that although short layovers on long-haul flights did not significantly change pilots' sleep patterns, they did result in increased levels of fatigue during and after the trip. Pilots with shorter layovers reported feeling more subjectively tired and finding it difficult to focus for prolonged periods than pilots with longer layovers. It's interesting to note that pilots with longer layovers—two sleep opportunities during local nighttime—showed improved attention capacity and a quicker recovery to pre-trip levels following just one recovery night. These results underscore the importance of layover duration in supporting pilot recovery and performance, with implications for scheduling policies aimed at mitigating fatigue-related risks in the aviation industry. The ideal stopover times for long-haul flights should be more carefully considered and investigated to improve pilot welfare and operational safety. (Roach et al., 2012, pp. 584-585).

Song, C., Guo, J., & Zhuang, J. (2020). Analyzing passengers' emotions following flight delays-a 2011–2019 case study on SKYTRAX comments. *Journal of Air Transport Management*, 89, 101903. <https://doi.org/10.1016/j.jairtraman.2020.101903>

The study looks at 24,165 SKYTRAX reviews from September 2011 to March 2019. It identifies passenger issues in the aviation industry through sentiment and co-occurrence analysis (Song et al, 2020, p. 1). This paper intends to use text mining to efficiently assess aircraft delay comments. It employs lexicon-based sentiment analysis to measure sentiments and co-occurrence analysis to uncover passenger reservations about various service components during flight delays in the aviation industry (Song et al, 2020, p. 2). SKYTRAX is an unofficial worldwide air transport rating organization that offers the only airline quality ranking service. (Song et al, 2020, p. 3). To preprocess the text, it underwent cleaning, segmentation, extended abbreviations, stop word removal, word correction, and lemmatization (Song et al, 2020, p. 3). To analyze the sentiment of reviews, the Vader and Pattern lexicon is used (Song et al, 2020, p. 4).

The results show that flight delays have an impact on the sentiment of passengers. Flight delays and compensation mechanisms play a significant role in this issue (Song et al, 2020, p. 6). The number of passengers dissatisfied with their airport experience was nearly five times higher than the number of satisfied customers (Song et al, 2020, p. 6). Sentiment analysis shows that travellers experience lesser service quality than expected, indicating a negative perception (Song et al, 2020, p. 7). The sentiment polarity of user comments correlates with their usefulness. Future work could focus on selecting comments with high sentiment polarity, analyzing user experience, and identifying problems (Song et al, 2020, p. 7).

H Van Landeghem, A Beuselinck (2002), Reducing passenger boarding time in airplanes: A simulation based approach, *European Journal of Operational Research*, 142(2), 294-308, [https://doi.org/10.1016/S0377-2217\(01\)00294-6](https://doi.org/10.1016/S0377-2217(01)00294-6).

Van Landeghem and Beuselinck's (2002) study investigates solutions to reduce passenger boarding time and increase flight turnaround times and on-time departures. Examining various boarding patterns and how they impact onboarding success is the primary goal of the study. Using simulation in Arena software, the researchers modelled the effects of different disruptions and boarding tactics on overall boarding time and passenger perception (Van Landeghem and Beuselinck, 2002, p. 8). To create and evaluate several boarding procedures, the study considered several variables, such as the amount of carry-on luggage, class sizes, and boarding sequences.

The study concludes that, in comparison to random boarding approaches, Individualized boarding calls can significantly reduce the total amount of time spent

boarding (Van Landeghem and Beuselinck, 2002, p. 15). The study emphasises how important it is to implement boarding protocols that effectively split passengers in succession in order to decrease interference and increase overall boarding efficiency. Because it clarifies how boarding policies impact passenger boarding time and experience, the research is important for airlines looking to enhance customer happiness and expedite their boarding processes.

Wang, Y., Cao, Y., Zhu, C., Wu, F., Hu, M., Dương, V., Watkins, M., Barzel, B., & Stanley, H. E. (2020). Universal patterns in passenger flight departure delays. *Scientific Reports*, 10(1). <https://doi.org/10.1038/s41598-020-62871-6>

The goal of this study is to identify the causes of propagation delays and establish common measurements to assess an airline's operational efficiency in reducing them (Wang et al, 2020, p. 1). The causes of passenger flight delays are classified into five categories. Carriers, extreme weather, national aviation system, security and late-arriving aircraft. These are called non-propagation factors. The propagated delay factor is known as the PD factor. PD occurs when the current flight is delayed due to the late arrival of the preceding flight with the same aircraft (Wang et al, 2020, p. 2). When the current departure delay is shorter than the previous one with the same aircraft, non-propagation factors may have a less significant effect (Wang et al, 2020, p. 3). Using large-scale data from BTS in the US, we identified two distinct patterns of propagated departure delays for passenger flights, classifying airlines with an SPL and those with ETSPL CCDF. Models 1 and 2 capture the two distinct delay propagation mechanisms in these groups. Although most airlines remained stable during this split, others have changed classification over time (EV, AS, DL) (see SM). This phenomenon needs a closer look (Wang et al, 2020, p. 8-9).

Wu, W., Wu, C., Feng, T., Zhang, H., & Qiu, S. (2018). Comparative analysis on propagation effects of flight delays: A case study of China Airlines. *Journal of Advanced Transportation*, 2018, 116–122. <https://doi.org/10.1155/2018/5236798>

Wu, Wu, Feng, Zhang, and Qiu (2018) investigate the propagation effects of flight delays caused by various factors such as airline operations, weather conditions, and air traffic control (ATC). In their study published in the *Journal of Advanced Transportation*, the authors utilize a copula function to analyze the interdependency among sequential flight delays (p. 116) and explore strategies to mitigate delay propagation. By examining actual data from an airline in the Asia Pacific region, they demonstrate how flight delays can propagate to downstream airports or airlines, affecting subsequent flights (p. 116). The study emphasizes the importance of understanding delay propagation dynamics and proposes scenarios to enhance flight schedule reliability through efficient buffer time allocation.

The research addresses the significant challenges posed by flight delays in airport management and scheduling, particularly in regions with high air travel demand such as China. Through a comprehensive analysis of delay factors and their impact on sequential flight delays (pp. 116-118), Wu et al. provide valuable insights into the complex nature of delay propagation. By employing copula-based methodologies (pp 118-120), the study offers a novel approach to quantifying the correlation between upstream and downstream flight delays, contributing to the development of effective strategies for managing delay propagation. Additionally, the study highlights the potential benefits of adjusting flight schedules with additional buffer time (pp. 120-122), underscoring the importance of proactive measures in mitigating the adverse effects of flight delays on airline operations and passenger satisfaction.

Zhang, D., & Klabjan, D. (2017). Optimization for gate re-assignment. *Transportation Research Part B: Methodological*, 95, 260–264. <https://doi.org/10.1016/j.trb.2016.11.006>

In their study on gate re-assignment within the airline sector, Zhang and Klabjan (2017) delve into the intricate challenge of efficiently managing gate operations amidst frequent disruptions such as flight delays and cancellations (p. 261). They underscore the crucial role of gate re-assignment planning in maintaining high service standards and passenger satisfaction. Gate controllers, responsible for adjusting gate assignments based on real-time flight schedules and existing plans, confront the daunting task of ensuring operational smoothness amid unforeseen events (Zhang & Klabjan, 2017, p. 261).

To tackle this complex issue, Zhang and Klabjan (2017) propose an innovative optimization methodology, recognizing the need for systematic approaches to address the gate re-assignment problem effectively (p. 261). Their multi-commodity network flow model treats each gate as a distinct entity and aircraft sequences as flows within the gate's network, providing a comprehensive representation of gate operations. This model integrates diverse objectives, such as minimizing flight delays, reducing the number of gate re-assignment operations, and mitigating the impact on connecting passengers (Zhang & Klabjan, 2017, p. 261). Additionally, the authors introduce heuristic algorithms to solve the problem efficiently, acknowledging the growing complexity of gate operations and the imperative for swift decision-making in response to disruptions (Zhang & Klabjan, 2017, p. 261). This research makes a significant contribution to the field by offering practical solutions to optimize gate operations and alleviate the effects of disruptions on airline service quality.

Appendix A1

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Appendix B: Surveys

We created a survey as part of our research to collect quantitative data and to identify the pain points from participants' experiences. A variety of sources were used to recruit participants, including email, social media, and in-person interactions. The informed consent was shown within the survey. Following the survey, participants were asked if they would like to participate in a 30-minute semi-structured interview. A total of 15 participants were selected for the interview.

[Informed consent](#) for participants

[Ravens Survey](#)

[Survey Responses and Analysis](#)

[Survey analysis document](#)

Section 1: Survey screener

Question 1: Have you experienced a delayed or missed connecting flight?

Answer	No of responses	% of responses
Yes	67	62.6%
No	39	36.4%
I Prefer not to say	1	0.9%

Question 2: What age group do you fall into?

Answer	Number	Percent
18 - 30 years	97	90.7%
31 - 40 years	9	8.4%
41 - 50 years	0	0.0%
51 years and above	1	0.9%
I prefer not to say	0	0.0%

Question 3: Were you travelling alone?

Answer	Number	Percent
Yes	90	15.0%
No	16	40.0%
I prefer not to say	1	0.9%

Question 4: Were you travelling internationally in economy class?

Answer	Number	Percent
Yes	88	82.2%
No	18	16.8%
I Prefer not to say	1	0.9%

Question 5: How many connecting flights did you have?

Answer	Number	Percent
1	40	37.4%
2	35	32.7%
3 or more	16	15.0%
No connecting flights	15	14.0%
I prefer not to say	1	0.9%

Section 2: Clarity and transparency

Question 1: Did you book your connecting flight as part of a single booking or separately?

Answer	Number	Percent
I booked my connecting flight as part of a single booking.	15	89.4%
I booked my connecting flight separately from the initial booking.	6	9.1%
Other	0	0.0%
I prefer not to say	1	1.5%

Question 2: If you faced a delayed or missed connecting flight, what was the reason?

Answer	Number	Percent
Bad weather	25	37.3%
Insufficient layover time	14	20.9%
Immigration process	15	22.4%
Personal reasons	5	7.5%
The reason was not apparent	14	20.9%

Other	11	16.5%
I prefer not to say	2	3.0%

Question 3: How satisfied were you with the airline's response to your delayed or missed connecting flight?

Answer	Number	Percent
Very dissatisfied	19	28.8%
Dissatisfied	18	27.3%
Neither dissatisfied nor satisfied	19	28.8%
Satisfied	8	12.1%
Very satisfied	2	3.0%

Question 4: To what extent do you agree with the statement; I received great service the last time I faced a flight disruption?

Answer	Number	Percent
Strongly disagree	19	28.4%
Somewhat disagree	21	31.3%
Neither agree nor disagree	16	23.9%
Somewhat agree	5	7.5%
Strongly agree	6	9.0%

Question 5: Did you receive any of the following when you faced a flight disruption without needing to ask the airline?

Answer	Number	Percent
Hotel Accommodation	10	17.9%
Meals	11	19.6%
Baggage handling services	11	19.6%
Monetary compensation	8	14.3%
Revised travel plan (Rebooking of flights)	17	30.4%

Vouchers and offers	11	19.6%
Other: _____	13	23.3%
I prefer not to say	5	8.9%

Section 3: Customer expectations and past experiences

Question 6: Did you know you are entitled to compensation when you face flight disruption due to an error made by the airline carrier?

Answer	Number	Percent
Yes	36	53.7%
No	30	44.8%
I prefer not to say	1	1.5%

Question 7: Have you asked for compensation due to a flight disruption in the past?

Answer	Number	Percent
Yes, I have	19	28.4%
No, I have not	47	70.1%
I prefer not to say	1	1.5%

Question 8: If you asked for compensation, what happened?

Answer	Number	Percent
I received compensation	15	22.4%
My request was denied	6	9.0%
I did not receive compensation	35	52.2%
I prefer not to say	11	16.4%

8a: If you received compensation for the delayed or missed connecting flight, in what form was it?

Answer	Number	Percent
Monetary compensation	5	33.3%

Vouchers or offers	8	53.3%
Meals and accommodation	1	6.7%
Other: _____	3	20.1%
I prefer not to say	0	0.0%

8b: On a scale of 1 to 5, how satisfied were you with the compensation you received?

Answer	Number	Percent
1 (Least satisfied)	3	20.0%
2	2	13.3%
3	5	33.3%
4	4	26.7%
5 (Most satisfied)	1	6.7%

Question 9: If your request was denied, what was the reason for the denial of compensation?

Answer	Number	Percent
_____	6	100.0%

Section 4: Accessibility of information

Question 10: To what extent do you agree that you were informed about the airlines' policies and practices for managing disrupted flights before the flight? (scale)

Answer	Number	Percent
Strongly disagree	28	41.8%
Disagree	13	19.4%
Neither agree nor disagree	19	28.4%
Agree	6	9.0%
Strongly agree	1	1.5%

Question 11: Have you read the Air Passenger Protection Regulations?

Answer	Number	Percent
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Yes, I have	13	19.4%
No, I have not	51	76.1%
I prefer not to say	3	4.5%

Question 12: If you have read the Air Passenger Protection Regulations, where did you find it?

Answer	Number	Percent
The official airline page	7	11.7%
The Canadian Transportation Agency (CTA)	4	6.7%
Family and friends	1	1.7%
At the airport	4	6.7%
I did not read or find the information	34	56.7%
Other: _____	4	6.5%
I prefer not to say	6	10.0%

Section 5: Wrap up questions

Question 13: Do you wish to enter the draw for one of four \$25 gift cards?

Answer	Number	Percent
Yes	49	73.1%
No	18	26.9%

Question 14: Would you like to participate in a 45-minute follow-up interview?

Answer	Number	Percent
Yes	21	31.3%
No	46	68.7%

If you answer Yes to any of the previous questions, please enter your email address

Email: _____

Appendix C – Subject Matter Expert (SME) interview materials

[Recruitment protocol for SMEs.](#)

Signed [Informed consent for SMEs.](#)

[Interview protocol for SMEs.](#)

[SME Interview Analysis -Miro](#)

Transcription notes

SME 1: Station operations coordinator (recording link [S_ID1 interview](#))

SME 2: Professor of Aircraft Maintenance course (recording link [S_ID2 interview](#))

SME 3: Professor of Aircraft Maintenance course (recording link [S_ID3 interview](#))

SME 4: Pilot (recording link [S_ID4 interview](#))

Appendix D – Study participant interview materials

[Participant recruitment protocol](#)

[Participant informed consent](#)

[Participant interview protocol](#)

[Participant interview analysis -Miro](#)

Transcription notes:

Participant 1: [P_ID1 folder](#)

Participant 2: [P_ID2 folder](#)

Participant 3: [P_ID3 folder](#)

Participant 4: [P_ID4 folder](#)

Participant 5: [P_ID5 folder](#)

Participant 6: [P_ID6 folder](#)

Participant 7: [P_ID7 folder](#)

Participant 8: [P_ID8 folder](#)

Participant 9: [P_ID9 folder](#)

Participant 10: [P_ID10 folder](#)

Participant 11: [P_ID11 folder](#)

Participant 12: [P_ID12 folder](#)

Thematic analysis:

[Affinity mapping](#)

[Interview \(Analysis\)](#)

[illegible][illegible]

Flight Delay Reason	Passenger Reaction
They fly regional The first regional flight on the east there was a maintenance issue	They got worried They were worried that they would be late for work and that they would miss their flight
Engineers had to be brought from Montreal The person there said that they had to call them that they were not there and they had to bring them to the airport and then they had to wait for them to arrive	They got angry They were angry that they were late and that they were not being told about the delay
The reason was changed to mechanical issues The reason was changed to mechanical issues and they had to wait for the flight to be rescheduled	They got angry They were angry that they were late and that they were not being told about the delay
Delay was due to technical issue The delay was due to a technical issue and they had to wait for the flight to be rescheduled	They got angry They were angry that they were late and that they were not being told about the delay
They said when they were informed that they could not board because of technical issues They said when they were informed that they could not board because of technical issues	They got angry They were angry that they were late and that they were not being told about the delay
Passenger felt that the reason for the weather delay was genuine Passenger felt that the reason for the weather delay was genuine	They got angry They were angry that they were late and that they were not being told about the delay
No reason was told to him, they just said something was wrong No reason was told to him, they just said something was wrong	They got angry They were angry that they were late and that they were not being told about the delay
They got delayed because of baggage handling They got delayed because of baggage handling	They got angry They were angry that they were late and that they were not being told about the delay
The participants flight arrived to the airport on the wrong day and they had to wait for the flight to be rescheduled The participants flight arrived to the airport on the wrong day and they had to wait for the flight to be rescheduled	They got angry They were angry that they were late and that they were not being told about the delay
Not even after 2 hours there was an emergency landing Not even after 2 hours there was an emergency landing	They got angry They were angry that they were late and that they were not being told about the delay
Second flight was delayed because of bad weather Second flight was delayed because of bad weather	They got angry They were angry that they were late and that they were not being told about the delay
"I'm afraid of this, this is the flight that I'm afraid of, they are going to be late" "I'm afraid of this, this is the flight that I'm afraid of, they are going to be late"	They got angry They were angry that they were late and that they were not being told about the delay
"Knowing nothing is better than knowing something concerning them" "Knowing nothing is better than knowing something concerning them"	They got angry They were angry that they were late and that they were not being told about the delay
The participants mentioned that connections to domestic airports will always have a delay The participants mentioned that connections to domestic airports will always have a delay	They got angry They were angry that they were late and that they were not being told about the delay
"I'm afraid of this, this is the flight that I'm afraid of, they are going to be late" "I'm afraid of this, this is the flight that I'm afraid of, they are going to be late"	They got angry They were angry that they were late and that they were not being told about the delay
"They can't be honest to me, this flight, if the flight is going to be late, they should just say so, they should just say so" "They can't be honest to me, this flight, if the flight is going to be late, they should just say so, they should just say so"	They got angry They were angry that they were late and that they were not being told about the delay
"We had a lot of trouble because they were trying to hide the fact that they were going to be late" "We had a lot of trouble because they were trying to hide the fact that they were going to be late"	They got angry They were angry that they were late and that they were not being told about the delay
Passenger mentioned the airline they chose to fly to almost always late Passenger mentioned the airline they chose to fly to almost always late	They got angry They were angry that they were late and that they were not being told about the delay

The participant wasn't even told they had rights because they weren't informed of such a document exists. yes no maybe green/white	It would have been better if the participant had been made aware of such a document. yes no maybe green/white	If the participant was made aware of the rights and the communication channels would have been effective. yes no maybe green/white	Participant straight up said "No". They never read it. yes no maybe green/white	They were not told and couldn't read the document. They were not informed of such a document exists. yes no maybe green/white
The participant never felt informed about rights yes no maybe green/white	No one mentioned about such rights to the participant. yes no maybe green/white	When asked about participant rights, the participant mentioned they are not sure about those are. yes no maybe green/white	The participant just assumed through the terms and conditions because there was no consent information yes no maybe green/white	"We don't think that we're going to need that anymore soon so we're just like, okay, just forget everything" yes no maybe green/white
Participant said they kind of thought. yes no maybe green/white	"I would say, yes, of utmost importance" yes no maybe green/white	he believe rights are not that much relevant to customers so they did not read it to rights. yes no maybe green/white	Participant said it's helpful but so of these three documents, one would have to be involved in regulatory. yes no maybe green/white	People don't really take that much time and go into reading it, they just do it on the day of the signing the document. yes no maybe green/white
"After going through this, like a survey, they just read that document, they didn't actually 'know' anything about 'exercising any right'." yes no maybe green/white	The participant only read the cancellation policy yes no maybe green/white			

[illegible]

100% Transparency Transparency between airlines and passengers Transparency means airlines need to give passengers the information they need to make informed choices. Key points: - Airlines - Passengers - Information	There is no communication gap between airlines and passengers. Key points: - Airlines - Passengers - Information	There should be a better system of communication between the two different airlines. Key points: - Airlines - Passengers - Information
Passenger and communication is the key and is most important. Key points: - Airlines - Passengers - Information	The information about the flight arrived early. Key points: - Airlines - Passengers - Information	Communication is vital throughout the entire flight and made passengers feel comfortable. Key points: - Airlines - Passengers - Information
		Clear Communication. Key points: - Airlines - Passengers - Information

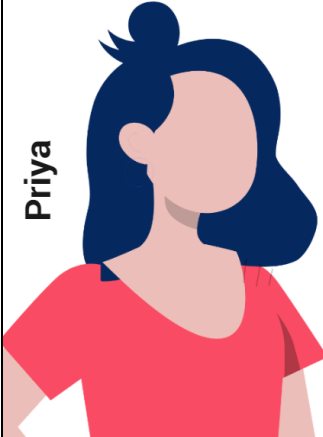
I received emails giving pop up messages to passengers button	Email notification would have good Even if they communicated via email, it couldn't replace it due to lack of network access button	The participant received email updates on the flight's status message
passenger expected emails. They would have preferred email updates. button	an email would be good 'an email with a detailed description or a detail of things going on would be really give us an assurance' button	

[illegible][illegible]

Appendix E: Research Design Artifacts

Personas

[Personas -Miro](#)



Priya

"I feel very anxious, yet I am pretty excited to start a new chapter of my life. I haven't travelled a lot, but I hope my journey goes well."

Demographics:

Age: 25
Profession: International Student
Country: India

Characteristics

- 😊 Excited and Enthusiastic
- 😬 Anxious and Curious
- 💻 Technologically Savvy

Goals

- 🎯 Ensure timely arrival in Canada to begin her higher education program without any delays.
- 🎯 Have a smooth and comfortable travel experience.
- 🎯 Get to Canada early because her orientation is in two days

Context

- **Travel:** Connecting flight from India to Canada for higher education.
- **Situation:** The flight was significantly delayed, and the airline provided poor communication and minimal support. She did not receive any accommodation, meals or coupons for the delay.

Pain Points

- **Travel-related anxiety and excitement** due to her first international trip
- **Worry about missing flight** and additional security procedures.
- **Fear of traveling alone** all by herself.



Jose

"I am so excited to be attend my brother's wedding! I hope my journey goes well, but just in case if any issue arises, I feel like I would be able to handle them pretty well by myself."

Demographics:

Age: 24
Profession: Former International Student
Now a job-seeker
Country: Spain

Characteristics

- 😬 Curious
- 😊 Patient
- 📞 Communicative

Goals

- 🎯 Affordable flight tickets to Spain
- 🎯 Booking with a well-known airline to have good flight experience.
- 🎯 Spend quality time with the family and friends back in Spain.

Context

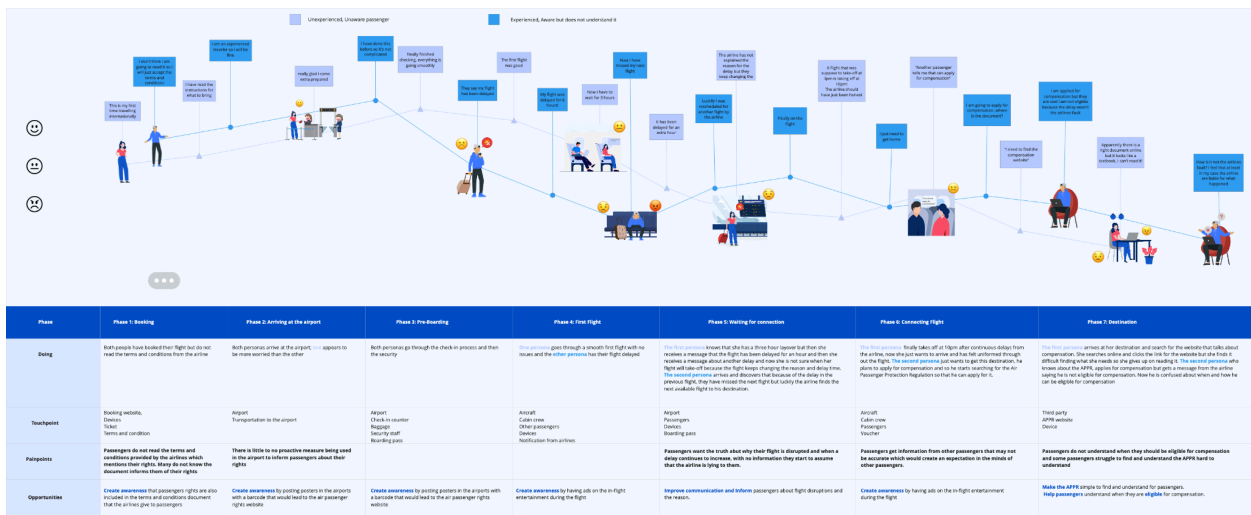
- **Situation:** He was very excited for his journey. Unfortunately, his first flight got delayed. When he reached the connecting airport, he was informed that his connecting flight has left without him. He was worried. From his past travels, he knew that he can ask for a compensation in such kind of situation. Beaming with hope, he asks for compensation from airlines. The airline informs him that he is not eligible for the compensation. Jose got irritated with the response and found the rights confusing.
- **Travel:** Flight from Canada to Spain with a connecting flight in Toronto, he is heading home for a break.

Pain Points

- **Excitement tempered by concerns** about potential travel issues.
- **Fear of missing flight and facing delays** during his travel.

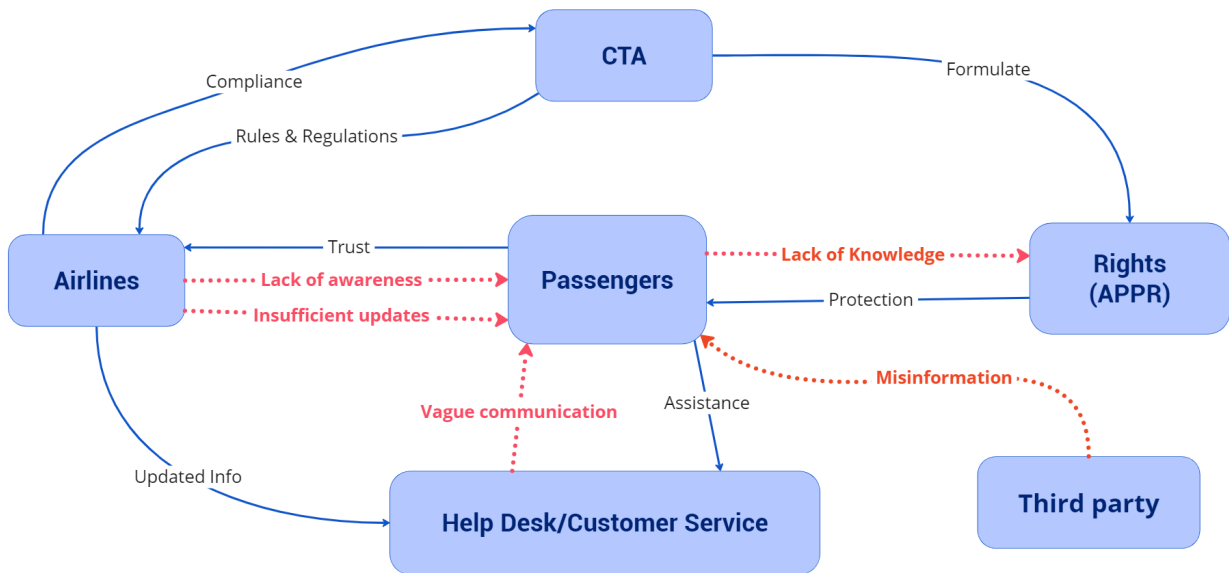
Journey Map

Go to the [Full Journey Map](#)



System Map

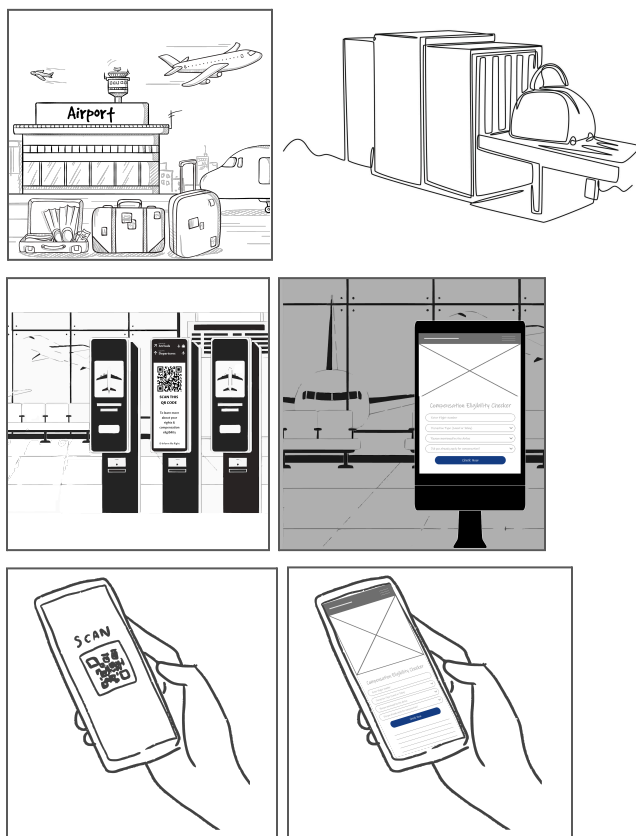
[System Map -Miro](#)



This is the system map that we used to illustrate the flow of information and responsibilities related to passenger rights:

The Canadian Transportation Agency (CTA) is responsible for formulating and implementing passenger rights to facilitate travelers in the event of service disruption and other irregularities. Moreover, the CTA has mandated that the airlines follow guidelines when advising passengers of their rights. Actually, passengers often get vague and incomplete information from airline help desks, thereby leaving them unaware and uninformed of their rights. This sometimes leaves information gaps, whereby passengers seek information through unofficial third-party sources, consequently leading to misinformation. It is quite important that airlines improve transparency and communication on passenger rights so more protection can be accorded to the passengers so that travelers are fully informed and empowered to act upon accurate and reliable information.

Storyboard



This storyboard should establish, through visual means, how air passengers facing delays in an airport can quickly and easily understand their rights to make their experience more knowledgeable and less stressful. The sequence of events in this storyboard describes the passage of a passenger who feels lost and anxious after hearing about a delayed flight but then finds clarity and solace through a digital tool specially built to help in such scenarios.

Initial Mock-ups

Inform Me Right

HomeAboutEligibility CheckAPPR

Helping Passengers know their **rights**!

Did you face a disruption and are confused what steps to take next? Or are you curious what rights do you have as a air traveler? We are here to help you!

Compensation Eligibility Checker

Read Simplified APPR

Inform Me Right

HomeAboutEligibility CheckAPPR

Understand **rights** as a **Passenger**!

We know rights could be hard to understand, so we decided to make it easier for you.

General

Delay, Cancellation and Denial of Boarding

Baggage

Advertising

Administrative Monetary Penalties

Transitional Provisions

Amendments to these Regulations

Consequential Amendments

Air Transportation Regulations

Compensation Eligibility Checker

Enter Flight Number

Disruption Type (Cancel or Delay)

Reason mentioned by the Airline

Did you already apply for compensation?

Check Now

Appendix F: Design Validation Materials

Validation Interviews and Protocol

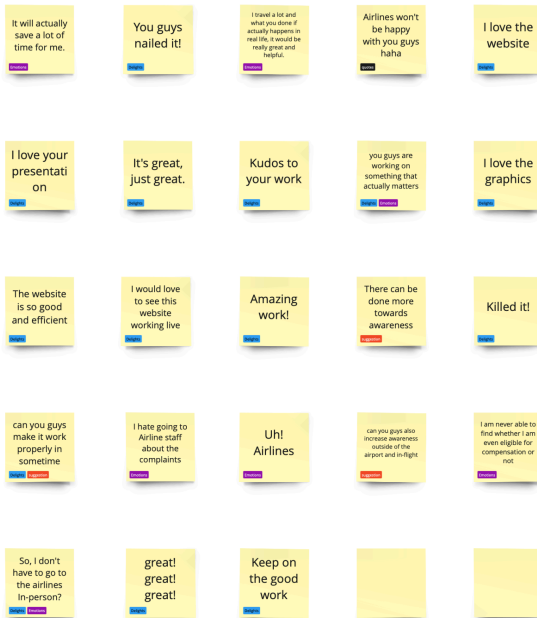
[Validation folder](#)

[Validation protocol](#)

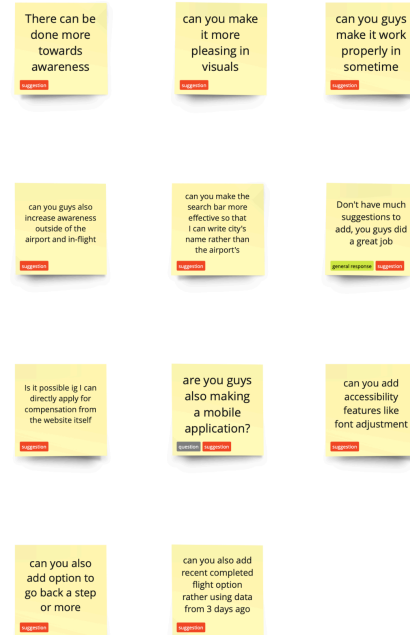
[Participants Responses -Miro](#)

[Suggestions -Miro](#)

Participants responses



Suggestions



Poster



Project Inform-Me-Right

Our research examines the factors contributing to the communication gap between airline services and passenger's expectations during flight disruptions, as well as the passengers' compensation eligibility. The impact of our research will help to improve awareness and knowledge about the Air Passenger Protection Regulations (APPR).

Team RAVENS

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ALGONQUIN COLLEGE | Human-Centred Design Lab



National Research Council Canada

Conseil national de recherches du Canada

Methods

107 Surveys **12** Interviews **4** SME's

Findings

"So it's like what needs to happen for me is to have the right to get the compensation."

"I would not tell the name of the company, but I will refrain using that airline."

"It's like so much to read through, it's like reading a textbook it would be better if important points were highlighted."

"an email with a detailed description or a detail of things of what's going on would have really given us an assurance."

Interventions

1 Raise Awareness

Raising awareness about the rights and regulation document



3 Make It Simple

Make the rights and regulations document more understandable for passengers, despite containing a lot of information



2 A Digital Platform

Creation of digital platform



4 Make It Clearer & Accessible.

Simplify the information regarding compensation eligibility, and make it more accessible for passengers



Final Prototype

Figma Prototype Link: [Figma](#)

Understand your Rights as a Passenger

The Air Passenger Protection Regulation (APPR) applies to **all** flights to, from, and within Canada. This includes connecting flights. If you experienced a problem, you must first contact your airline to resolve it. If that does not work, you can file a complaint with us. The CTA processes complaints and settles disputes between travelers and airlines.

[Compensation Eligibility](#)[Read the Full APPR](#)[Make a complaint](#)

What is your situation?

Select the option that applies to you

- ☐ My flight was delayed or cancelled
- ☐ I missed my connecting flight
- ☐ I lost my baggage or my baggage was damaged
- ☐ I was denied boarding
- ☐ I want clear communication

[Enter](#)

My flight was delayed or cancelled

Jump to a topic

- ▶ Your Rights
- ▶ During a flight disruption
- ▶ Compensation
- ▶ Compensation level
- ▶ Communication
- ▶ Assistance
- ▶ Rebooking and refunds



Conclusion for Case Study Recommendation

Key Recommendations:

- **Post-disruption Communication:** Users expressed a desire for consistent and accurate communication following flight disruptions. It's suggested that this is a critical area where the system could improve by providing timely and clear updates.
- **Human Interaction:** The importance of maintaining a human touch in communication was emphasized. Users prefer a balance between automated responses and access to human support.
- **Proactive Information:** Users appreciate when airlines provide proactive updates about delays or disruptions, rather than waiting until the last minute.
- **Clarity and Transparency:** Participants valued transparency in communication, particularly when it came to understanding the reasons behind disruptions and the expected outcomes.

User Feedback Highlights:

- **Timeliness:** Users often feel that communication regarding delays is too slow, causing frustration.
- **Accuracy:** There is a significant need for accurate information that doesn't change frequently or contradict itself.
- **Personalization:** Users prefer communication that is personalized and relevant to their specific situation.
- **Multiple Channels:** Offering updates through multiple channels (email, SMS, app notifications) is important, but consistency across these channels is crucial.

Suggestions for the Website:

- **Centralized Information Hub:** The website could serve as a centralized hub for all updates related to disruptions, accessible via mobile devices.
- **User-Friendly Interface:** Ensure that the interface is intuitive and accessible, particularly for users under stress.
- **Multilingual Support:** Consider offering information in multiple languages to accommodate international travelers.

These findings and recommendations could serve as a valuable foundation for improving the communication aspects of your website, ensuring that it aligns with user needs and preferences.